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Modification and Improvement of Automated Cake Frosting Machine

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Final Year Project Report

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Declaration

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Abstract

Automated cake frosting systems have the potential to improve consistency, reduce labour requirements, and increase productivity within cake production processes. However, a previously developed automated cake frosting machine encountered several limitations, including insufficient extrusion force, frosting inconsistency, nozzle clogging, limited synchronization between machine subsystems, and an inability to effectively frost cake side walls. These limitations reduced overall machine performance and restricted its practical application.

This project presents the modification and improvement of the existing automated cake frosting machine through the redesign of its mechanical, electrical, and software subsystems. The redesign was driven by the selection of buttercream frosting as the target frosting material, whose rheological characteristics required improved force generation, controlled extrusion, and synchronized motion control.

The developed system incorporates a moving-gantry Cartesian motion platform, a motorized rotary cake table, a syringe-based extrusion mechanism, an ESP32-based control system, and a wireless web-enabled operator interface. In addition, cake mapping algorithms were developed to automatically determine cake dimensions, while automated frosting routines were implemented to support circular frosting, side-wall frosting, leveling operations, and future text piping functionality.

The hopper-based extrusion mechanism used in the previous machine was replaced with a lead screw driven syringe system to improve frosting delivery and enable the extrusion of higher-viscosity buttercream frosting. A rotary platform was integrated to support synchronized cake rotation, while custom software algorithms were developed to coordinate motion control and frosting deposition.

The modified machine was successfully fabricated, integrated, and tested. Experimental evaluation demonstrated successful automated circular top-surface frosting using buttercream frosting and confirmed effective synchronization between motion control, rotary positioning, and frosting extrusion. The results indicate that the implemented modifications significantly improved machine functionality and addressed several limitations observed in the previous design.

The project demonstrates the feasibility of enhancing existing automated cake frosting systems through improved extrusion, motion control, and software integration, while providing a foundation for future development of fully automated cake decoration technologies.

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Symbols

τ	Shear stress	[Pa]
τ_y	Yield stress	[Pa]
$\dot{\gamma}$	Shear rate	[s ⁻¹]
$\delta_{\max}$	Maximum deflection	[mm]
Φ	Flow Multiplier	[-]
ϕ	Air volume fraction	[-]
η	Mechanical efficiency	[-]
ρ	Density	[kg/m ³]
K	Consistency index	[Pa·s ^{<i>n</i>}]
n	Flow behavior index	[-]
P	Applied load	[N]
L	Length / Span	[mm]
E	Young's Modulus	[GPa]
I	Area Moment of Inertia	[mm ⁴]
Q	Volumetric flow rate	[mm ³ /s]
V	Volume	[mL]
F	Force / Thrust	[N]
T	Torque	[Nm]
D	Diameter	[mm]
H	Height	[mm]
t	Thickness	[mm]

Abbreviations

CNC	Computer Numerical Control
ESP32	Extended Stack Processor 32-bit
NEMA	National Electrical Manufacturers Association
CAD	Computer-Aided Design
MTC	Manual Tool Change
ATC	Automated Tool Change
H-B	Herschel-Bulkley
ID	Internal Diameter
KES	Kenyan Shillings

Chapter 1

CHAPTER ONE

INTRODUCTION

1.1 Background

The food processing industry has increasingly adopted automation technologies to improve production efficiency, product consistency, and operational reliability. Automated systems are widely used in food manufacturing processes such as mixing, filling, packaging, sorting, and quality inspection. The integration of mechatronic systems into food production has enabled manufacturers to reduce dependence on manual labour while improving process repeatability and throughput.

Cake decoration represents an important stage in the production of baked products. Frosting serves both functional and aesthetic purposes by enhancing product appearance, providing additional flavour, and improving consumer appeal. Traditionally, frosting operations are performed manually using spatulas, piping bags, rotating cake stands, and other hand-operated tools[2]. The quality of the final product often depends on the skill and experience of the operator.

Although manual frosting techniques are widely practiced, they present several challenges. Achieving uniform frosting thickness, smooth surface finishes, and repeatable decorative patterns requires significant operator skill and can be time-consuming. Variations in frosting application may result in inconsistent product quality, particularly when large numbers of cakes must be produced within limited timeframes.

The automation of cake frosting presents unique engineering challenges compared to many conventional manufacturing processes. Unlike rigid materials, cake frosting behaves as a complex food material whose flow characteristics vary under applied force. The frosting must be capable of flowing through a delivery system during deposition while simultaneously retaining its shape after placement on the cake surface[7]. These characteristics require careful consideration of frosting delivery mechanisms, motion control

systems, structural stability, and process synchronization.

Recent developments in computer numerical control (CNC) systems, embedded control platforms, additive manufacturing technologies, and food deposition systems have created opportunities for the development of automated frosting machines capable of performing controlled and repeatable frosting operations[4,14]. Such systems may improve frosting consistency, reduce labour requirements, and enhance production efficiency while maintaining acceptable product quality.

This project focuses on the design and fabrication of an automated cake frosting machine capable of depositing frosting onto both the top surface and sidewalls of cylindrical cakes through the coordinated operation of a motion control system, rotary positioning platform, and controlled frosting extrusion mechanism.

1.2 Problem Statement

Cake frosting remains a largely manual operation in many bakery and confectionery environments. The process often relies on operator experience to achieve acceptable frosting quality, resulting in variations in frosting thickness, surface smoothness, and overall appearance. Manual frosting methods may also be labour intensive and time consuming, particularly when producing multiple cakes with consistent finishing requirements.

Previous attempts at automating cake frosting have encountered several challenges related to frosting delivery, motion synchronization, and process repeatability. In particular, the deposition of frosting onto cake sidewalls requires accurate coordination between frosting extrusion and rotational cake movement. Insufficient extrusion force, frosting flow inconsistency, nozzle clogging, and poor synchronization can negatively affect frosting quality and limit machine performance.

A previous automated cake frosting machine developed within the department utilized a hopper-based frosting delivery system and a three-axis motion arrangement. Although the system demonstrated the feasibility of automated frosting application, limitations were observed in frosting consistency, extrusion capability, sidewall frosting performance, and rotary-axis synchronization. These limitations restricted the machine's ability to achieve reliable and repeatable frosting deposition.

There is therefore a need for an improved automated cake frosting system capable

of providing controlled frosting extrusion, synchronized multi-axis motion, enhanced repeatability, and improved frosting coverage on both horizontal and cylindrical cake surfaces.

1.3 Main objective

To design and fabricate an automated cake frosting machine capable of achieving controlled and repeatable frosting deposition through synchronized multi-axis motion and syringe-based extrusion.

1.4 Specific objectives

1. To investigate the characteristics and engineering requirements associated with automated cake frosting.
2. To design a motion system capable of positioning a frosting nozzle over the required cake workspace.
3. To design and implement a rotary positioning mechanism for coordinated sidewall frosting operations.
4. To develop a syringe-based frosting extrusion system capable of delivering controlled frosting flow.
5. To design the electrical, control, and power subsystems required for machine operation.
6. To fabricate and integrate the mechanical, electrical, and control components of the proposed machine.
7. To evaluate the capability of the developed system to perform controlled frosting deposition.

1.5 Scope

This project focuses on the design, fabrication, and integration of an automated cake frosting machine intended for frosting application on cylindrical cakes. The work includes the development of a moving gantry Cartesian motion system, a rotary cake platform, a syringe-based frosting extrusion mechanism, electrical power and control systems, embedded software implementation, and custom fabricated components required for machine operation.

The project encompasses computer-aided design (CAD), component selection, subsystem integration, fabrication, and functional testing of the machine. The study primarily considers frosting deposition requirements and the engineering systems necessary to achieve controlled frosting application.

The project does not include commercial product development, industrial-scale production optimization, food safety certification, or extensive rheological characterization of frosting materials under laboratory conditions.

1.6 Significance of the Study

The development of automated food processing equipment contributes to the broader adoption of automation technologies within the food manufacturing sector. This project demonstrates the application of mechatronic design principles, motion control systems, and controlled material deposition techniques to a practical food-processing problem.

The study provides insight into the challenges associated with automating frosting deposition and contributes knowledge relating to frosting extrusion, synchronized motion control, and food-material handling. The developed machine may serve as a foundation for future research involving automated cake decoration, food deposition systems, and culinary automation technologies.

The project additionally provides an improved frosting system that addresses limitations identified in previous machine implementations through enhanced extrusion capability, improved structural design, synchronized rotary motion, and programmable multi-axis control. The resulting work contributes to the advancement of automated frosting technologies while providing valuable experience in the integration of mechanical, elec-

trical, electronic, and software engineering systems.

Chapter 2

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

The automation of food processing operations has become increasingly important as manufacturers seek to improve production efficiency, product consistency, and operational reliability. Advances in mechatronic systems, embedded control technologies, computer numerical control (CNC) systems, and automated material deposition have enabled the development of machines capable of performing tasks that were traditionally completed manually. One such task is cake frosting, which requires controlled material deposition, accurate positioning, and consistent surface finishing.

Cake frosting differs from many conventional manufacturing processes because the material being handled exhibits complex flow behaviour. Frosting must be capable of flowing through a delivery system during application while simultaneously retaining its shape after deposition[6]. Achieving this balance requires careful consideration of frosting rheology, extrusion methods, motion control systems, structural design, and process synchronization.

A successful automated cake frosting machine therefore requires the integration of several engineering disciplines including mechanical design, motion control, embedded systems, manufacturing methods, and food material handling. Understanding the principles governing each of these areas is essential for selecting appropriate design solutions and developing a machine capable of producing consistent frosting results[15].

This chapter reviews literature relevant to automated cake frosting systems, frosting material behaviour, food deposition mechanisms, motion systems, rotary positioning systems, motion transmission technologies, manufacturing methods, embedded control systems, and challenges associated with automated food deposition. The reviewed literature establishes the theoretical foundation upon which the design methodology and

engineering decisions presented in subsequent chapters are based.

2.2 Automated Cake Frosting Systems

Cake frosting is an essential finishing process in cake production and decoration. Frosting improves product appearance, enhances flavour, and contributes significantly to consumer perception of product quality. Depending on production scale and available technology, frosting operations may be performed manually, semi-automatically, or through fully automated systems.

The level of automation employed directly influences frosting consistency, production rate, labour requirements, and repeatability. As a result, numerous approaches have been developed to improve frosting application efficiency while maintaining acceptable product quality.

2.2.1 Manual Cake Frosting

Manual frosting remains the most widely used method of cake decoration, particularly in small bakeries, home baking environments, and custom cake production. In this approach[13], frosting is applied using hand-operated tools such as spatulas, scrapers, piping bags, and rotating cake stands.

The quality of manual frosting largely depends on the skill and experience of the operator. Experienced decorators are capable of producing visually appealing finishes and intricate decorative patterns. However, manual frosting presents several limitations.

- Inconsistent frosting thickness,
- Operator-dependent quality,
- Lengthy production times,
- Physical fatigue during repetitive operations,
- Limited repeatability between products.

Maintaining uniform frosting coverage on both horizontal and vertical cake surfaces can be particularly challenging when multiple cakes must be decorated within a short pe-

riod. Consequently, manual frosting becomes less suitable for applications requiring high production rates or consistent product standardization.



Figure 2.1: Manual Cake Frosting

2.2.2 Semi-Automated Cake Frosting Systems

Semi-automated frosting systems combine mechanical assistance with operator supervision. These systems typically automate specific portions of the frosting process while still requiring human intervention for setup, control, or finishing operations[10].

- Motorized cake turntables,
- Automated Frosting levellers,
- Assisted piping devices.

In many semi-automated systems, frosting delivery may be mechanized while nozzle positioning remains under operator control. Alternatively, cake rotation may be automated while frosting is applied manually.

The primary advantage of semi-automated systems is the reduction of physical effort required during frosting operations. These systems may also improve frosting uniformity and reduce production time when compared with fully manual methods[4].

However, because operator involvement remains necessary, product quality and process repeatability continue to depend on user skill. Variations in frosting thickness, deposition rate, and surface finish may still occur between production cycles.

Semi-automated systems therefore represent an intermediate stage between traditional frosting methods and fully autonomous frosting technologies.



Figure 2.2: Automated Frosting Leveller

2.2.3 Fully Automated Cake Frosting Systems

Fully automated cake frosting systems utilize programmable control systems to perform frosting operations with minimal operator intervention. These systems integrate motion control mechanisms, frosting delivery systems, embedded controllers, and software algorithms to coordinate frosting application automatically.

Unlike manual and semi-automated approaches, fully automated systems rely on controlled machine motion to position the frosting nozzle relative to the cake surface while simultaneously regulating frosting flow. Through coordinated operation of these subsys-

tems, frosting can be applied consistently across multiple products[9].

Modern automated frosting systems may employ Cartesian motion systems, robotic manipulators, rotary cake platforms, or hybrid motion architectures depending on the intended application. Some systems position the frosting nozzle around a stationary cake, while others rotate the cake beneath a fixed nozzle. More advanced systems combine both rotational and translational motion to achieve continuous frosting deposition on multiple cake surfaces.

The effectiveness of these systems depends heavily on the synchronization of several process variables, including:

- Frosting flow rate,
- Nozzle position,
- Cake rotation,
- Deposition speed,
- Frosting layer thickness.

Poor synchronization between these variables can result in frosting buildup, inconsistent coverage, surface defects, and reduced product quality. Consequently, motion control and extrusion control are critical considerations in automated frosting system design.

Looking toward future developments, automated frosting systems are expected to incorporate increasingly intelligent control algorithms, adaptive deposition strategies, machine vision technologies, and programmable decorative capabilities. Such advancements may enable machines to automatically determine cake geometry, adjust frosting parameters in real time, and generate customized decorative patterns with minimal user input[3,8].

2.2.4 Review of Previous Automated Cake Frosting Project

A previous automated cake frosting machine was developed within the department as an undergraduate engineering project. The machine demonstrated the feasibility of applying automation principles to cake frosting and provided a foundation upon which subsequent improvements could be developed.

The previous design utilized a three-axis configuration consisting of:

- An X-axis positioning system,
- A Z-axis positioning system,
- A rotary cake platform.

Frosting was supplied using a hopper-based delivery arrangement incorporating a rubber piston mechanism. Whipped cream was selected as the frosting material due to its relatively low viscosity and ease of extrusion.

Although the machine successfully demonstrated automated frosting application, several limitations were identified during operation.

One major challenge involved frosting delivery. The extrusion system experienced difficulties generating sufficient force to maintain consistent frosting flow, particularly when frosting properties varied[6]. This limitation contributed to inconsistent frosting deposition and reduced process reliability.

The system also experienced nozzle clogging and frosting flow interruptions during operation. Such interruptions negatively affected frosting quality and introduced variations in frosting thickness.

Another limitation involved side-wall frosting capability. The machine configuration primarily supported frosting operations on limited cake surfaces and lacked the coordinated motion required for effective frosting deposition around the cylindrical side walls of the cake.

Rotary motion synchronization also presented challenges. The rotational movement of the cake platform was not fully coordinated with frosting deposition and nozzle positioning, resulting in reduced repeatability and inconsistent frosting coverage.

The identified limitations can be summarized as:

- Insufficient extrusion force,
- Frosting inconsistency,
- Nozzle clogging,
- Limited side-wall frosting capability,
- Poor rotational synchronization,
- Reduced repeatability.

These observations highlighted the need for a redesigned automated cake frosting machine incorporating improved extrusion force generation, enhanced motion synchronization, greater structural stability, and the capability to apply frosting consistently to both the top and side surfaces of cylindrical cakes.

The limitations identified in the previous system directly informed the design decisions adopted in the present study and provided the motivation for the development of an improved automated cake frosting machine.

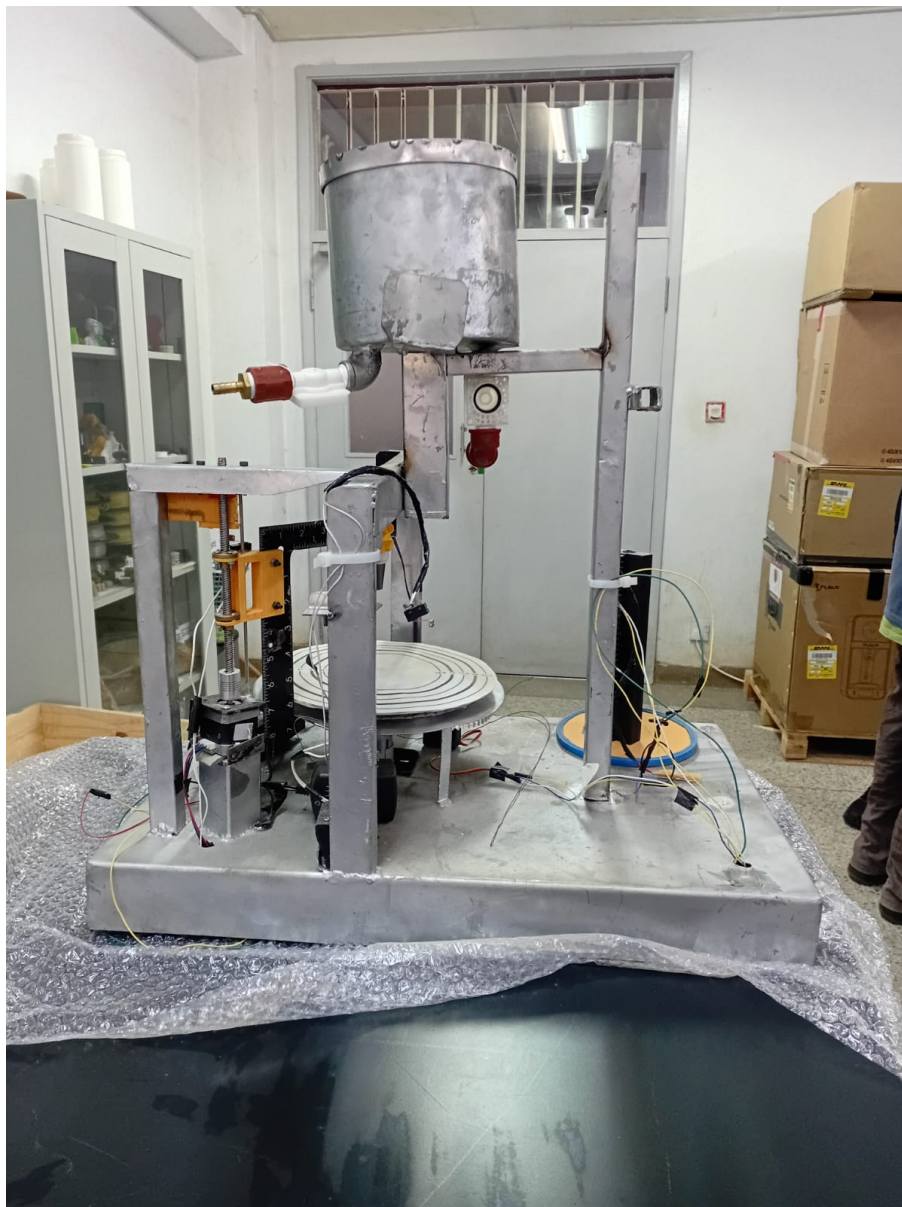


Figure 2.3: Previous Project Side View

2.3 Frosting Materials and Rheological Behaviour

The successful automation of cake frosting depends not only on machine design but also on the behaviour of the frosting material itself. Unlike rigid materials that retain a fixed shape under loading[1], frosting exhibits complex flow characteristics that influence how it is transported, extruded, deposited, and shaped during application. Understanding these material properties is therefore essential when designing automated frosting equipment.

The study of how materials deform and flow under applied forces is known as rheology. Rheological behaviour determines the amount of force required to move a material, the rate at which it flows through an extrusion system, and its ability to maintain shape after deposition. In automated frosting applications, rheology directly affects nozzle design, extrusion force requirements, motion synchronization, and final frosting quality.

2.3.1 Frosting as a Food Material

Cake frosting is a semi-solid food material used to coat, decorate, and enhance baked products. Depending on its composition, frosting may contain varying proportions of fats, sugars, water, air, and stabilizing ingredients[6]. These constituents influence both the texture and flow behaviour of the material.

An ideal frosting material for automated deposition must satisfy two seemingly contradictory requirements. First, it must be capable of flowing through a nozzle when sufficient force is applied. Second, it must retain its shape after deposition without spreading excessively or collapsing under its own weight. Achieving this balance is one of the primary challenges in automated frosting systems.

Unlike simple liquids, frosting does not behave uniformly under all loading conditions. Its resistance to flow changes depending on the applied force, deformation rate, temperature, and composition. As a result, frosting is generally classified as a complex non-Newtonian material.

2.3.2 Newtonian and Non-Newtonian Fluids

Materials may be broadly classified according to how their viscosity changes when subjected to deformation.

Newtonian fluids exhibit a constant viscosity regardless of the applied shear rate. This means that their resistance to flow remains unchanged whether they are moving slowly or rapidly. Common examples include water, vegetable oil, and many low-viscosity liquids.

In contrast, non-Newtonian materials exhibit viscosities that change under applied stress or deformation. Their flow behaviour depends on the magnitude and rate of the applied force[9,10].

A useful way to visualize rheological behaviour is to consider a spectrum between solid-like and liquid-like materials.

At one end of the spectrum are low-viscosity liquids such as water, which flow readily with very little applied force. Slightly more resistant materials such as syrups and sauces occupy an intermediate region. Frosting materials lie further toward the solid-like end of the spectrum, where significant force may be required before flow begins. Materials such as toothpaste and peanut butter exhibit similar behaviour. At the extreme end are materials such as cornstarch suspensions (commonly known as oobleck), which may behave like liquids under some conditions and solid-like materials under others.

This spectrum demonstrates that many food materials cannot be adequately described as simply liquids or solids. Instead, they exhibit characteristics of both depending on the loading conditions.

2.3.3 Shear-Dependent Behaviour of Frosting Materials

Many frosting materials belong to a class of non-Newtonian fluids whose apparent viscosity changes when force or motion is applied.

Some materials become easier to flow as the applied force increases, while others become more resistant to flow. These behaviours are commonly referred to as shear thinning and shear thickening[12].

Shear Thinning Behaviour

Shear-thinning materials become less viscous as the rate of deformation increases. In practical terms, the material flows more easily when subjected to greater force or motion.

This behaviour can be observed during manual cake decoration. Frosting may appear firm when resting in a container but spread smoothly when manipulated using a spatula or extruded through a piping nozzle. Once movement begins, less force is generally required to maintain flow[11].

Shear thinning is advantageous in cake decoration because it allows frosting to be deposited easily while still retaining its shape after application.

Shear Thickening Behaviour

Shear-thickening materials exhibit the opposite behaviour. Their apparent viscosity increases as the rate of deformation increases, causing the material to become more resistant to flow when subjected to higher forces.

A common example is a suspension of cornstarch and water, often known as oobleck. Under slow movement it behaves like a liquid, while rapid impact or deformation causes it to behave more like a solid[4,8].

Although shear thickening is generally undesirable for frosting applications, it provides a useful comparison for understanding the behaviour of non-Newtonian materials.

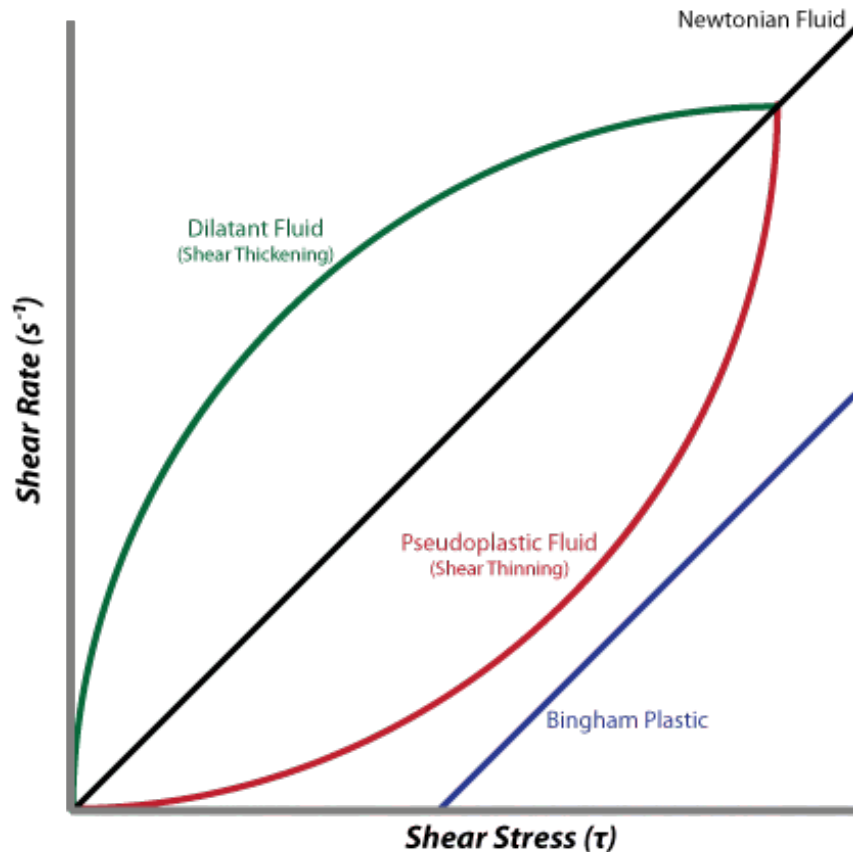


Figure 2.4: Graph of Rheology of Different Materials

2.3.4 Common Frosting Materials Used in Cake Decoration

Several frosting formulations are commonly used in cake decoration applications. Each possesses distinct rheological properties that influence its suitability for automated deposition.

Whipped Cream

Whipped cream consists primarily of cream that has been aerated to incorporate air bubbles. The resulting material is relatively light and possesses lower viscosity than many other frosting types. Because of its soft texture and reduced extrusion force requirements, whipped cream is often used in introductory frosting systems[14]. However, it exhibits limited structural stability and may deform under prolonged loading or elevated temperatures.



Figure 2.5: Whipped cream

Buttercream

Buttercream is one of the most widely used cake frosting materials. It typically consists of butter, sugar, and other ingredients combined to produce a smooth and stable frosting. Compared to whipped cream, buttercream generally exhibits higher viscosity, greater yield stress, and improved shape retention. These properties make it attractive for decorative applications but also increase the force required for extrusion[1].

Cream Cheese Frosting

Cream cheese frosting possesses a smooth texture and moderate viscosity. Its rheological behaviour lies between that of whipped cream and buttercream. The material provides good spreadability but may exhibit sensitivity to temperature variations[5].

Royal Icing

Royal icing is formulated using powdered sugar and egg whites or equivalent ingredients. Once deposited, it can harden significantly, making it suitable for decorative piping and detailed designs. Its flow characteristics differ substantially from softer frosting materials and require careful control during extrusion.

Ganache Ganache is typically produced from chocolate and cream mixtures. Depending on formulation and temperature, ganache may range from a relatively fluid material to a highly viscous frosting. Its rheological behaviour is strongly influenced by composition and environmental conditions.



Figure 2.6: Ganache

The diversity of frosting materials demonstrates that automated frosting systems must be capable of accommodating materials with varying viscosities, yield stresses, and flow characteristics.

2.3.5 Implications for Automated Frosting Systems

The rheological characteristics of frosting materials have significant implications for machine design. The presence of shear thinning behaviour and yield stress means that frosting cannot be handled using the same approaches employed for low-viscosity liquids.

Automated frosting systems must be capable of generating sufficient force to initiate material flow while simultaneously maintaining controlled deposition during operation. Excessive force may lead to over-extrusion[11],12, while insufficient force may result in inconsistent frosting delivery or nozzle clogging.

Furthermore, because frosting viscosity changes during flow, the extrusion system must operate in coordination with machine motion to maintain consistent frosting thickness. Variations in nozzle speed, cake rotation speed, or extrusion rate can directly affect frosting quality and surface finish.

The rheological behaviour of frosting materials therefore necessitates the use of controlled extrusion mechanisms, appropriate nozzle designs, and synchronized motion control strategies. These requirements form the basis for the selection and evaluation of frosting deposition systems discussed in the following section.

2.4 Extrusion Systems for Food Deposition

Food deposition systems are widely used in the food processing industry to transport, dispense, and accurately place food materials onto a product or surface. Applications include bakery filling, icing and frosting deposition, chocolate dispensing, sauce application, confectionery production, and emerging food-printing technologies.

The selection of an appropriate deposition system depends largely on the rheological properties of the material being handled. Low-viscosity liquids may be dispensed using relatively simple systems[8], whereas highly viscous materials require mechanisms capable of generating greater pressure and more precise flow control. Since frosting materials exhibit yield stress and non-Newtonian behaviour, the extrusion mechanism plays a critical role in determining deposition consistency and overall system performance.

Several extrusion methods have been developed for food deposition applications, each offering distinct advantages and limitations.

2.4.1 Gravity-Fed Systems

Gravity-fed systems rely on the weight of the material itself to generate flow through a nozzle or outlet. In these systems, frosting is stored within a container positioned above the dispensing point[9], allowing gravitational forces to drive material movement.

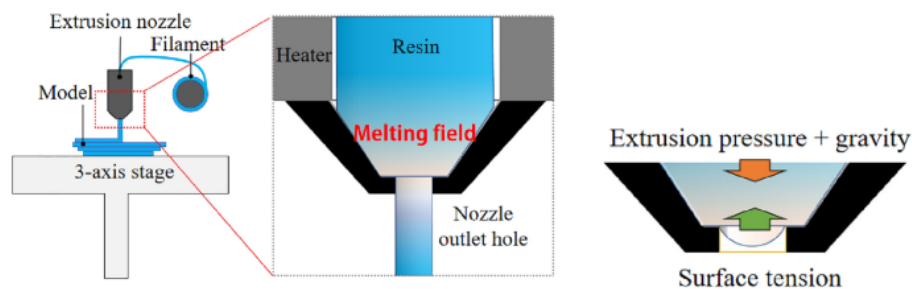
Gravity-fed systems are among the simplest deposition methods and are commonly used for low-viscosity liquids and semi-liquid food products.

Advantages include:

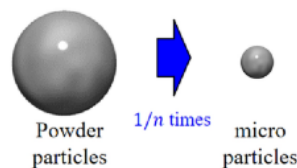
- Simple construction,
- Low cost,
- Minimal power requirements,
- Ease of operation.

However, gravity-fed systems present several limitations when handling frosting materials. The available driving pressure is limited by the height of the material column and decreases as the container empties. Consequently, flow rates may vary during operation, resulting in inconsistent deposition.

In addition, highly viscous frosting materials may not flow reliably under gravitational forces alone, making gravity-fed systems unsuitable for many cake frosting applications.



(a) Relationship between the surface tension and gravity force in the extrusion nozzle.



(b) Concept of moving $1/n$ times size particles.

Figure 2.7: Gravity-fed systems

2.4.2 Auger-Based Extrusion Systems

Auger extrusion systems utilize a rotating screw mechanism to transport material toward a dispensing nozzle. As the screw rotates, frosting is conveyed through a chamber and forced through the outlet.

Auger systems are widely used in food processing applications involving doughs, pastes, and other semi-solid materials.

Advantages include:

- Continuous material delivery,
- Suitability for relatively viscous materials,
- Good throughput capability,
- Compatibility with large-volume applications.

Despite these advantages, auger systems present several challenges for frosting deposition. The rotational action of the screw can introduce variations in flow rate and may alter the texture of aerated frosting materials. Cleaning and sanitation can also become more complex due to the presence of internal screw components.

For applications requiring highly accurate volumetric deposition, auger systems may offer less precision than positive mechanical displacement methods.

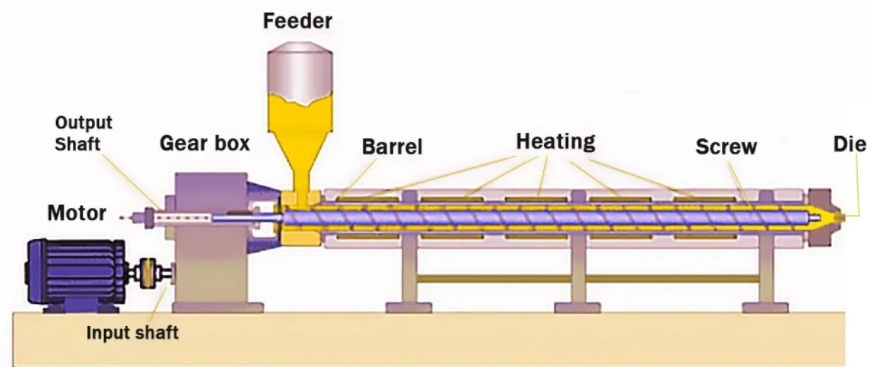


Figure 2.8: Auger-based extrusion system

2.4.3 Pneumatic Extrusion Systems

Pneumatic extrusion systems utilize compressed air to apply pressure to a material reservoir. The resulting pressure forces the frosting through a nozzle and onto the target surface[8,20].

These systems are widely employed in industrial dispensing applications because they provide relatively smooth material flow and can be adapted for various food products.

Advantages include:

- Smooth extrusion,
- Relatively simple mechanical design,
- Adjustable pressure control,
- Suitability for continuous operation.

However, pneumatic systems require external compressed air equipment and pressure regulation components. Variations in material viscosity may also influence flow behaviour, making precise volumetric control more challenging. Furthermore, compressibility of the air supply can introduce delays between control inputs and actual material flow.

For highly accurate frosting deposition, these limitations may reduce process repeatability.

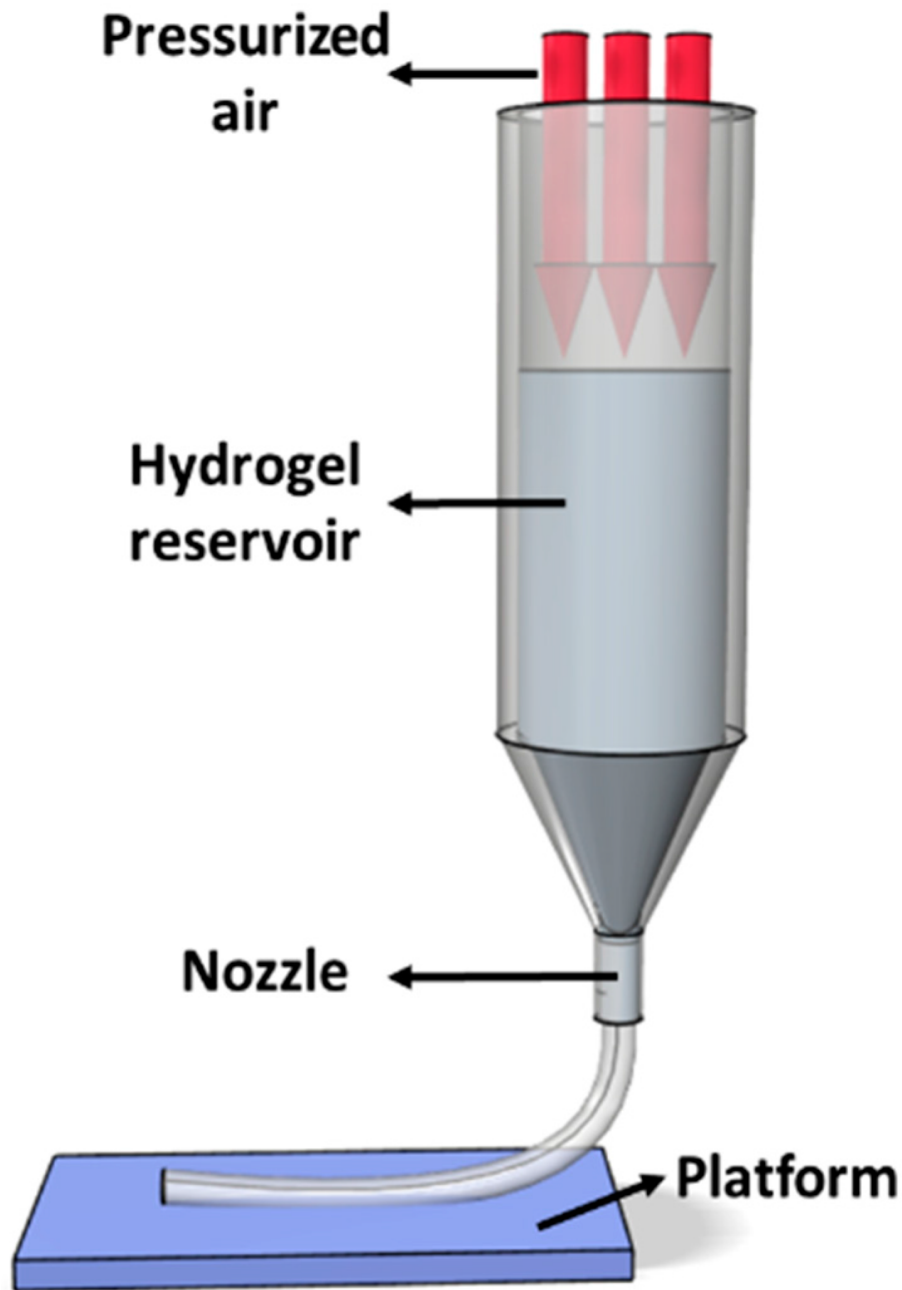


Figure 2.9: Pneumatic extrusion system

2.4.4 Syringe-Based Extrusion Systems

Syringe-based extrusion systems utilize a piston moving within a cylindrical chamber to force material through a dispensing nozzle. The piston may be driven manually or through mechanical actuation using lead screws, motors, pneumatic cylinders, or other force-generating mechanisms[16].

The syringe principle is fundamentally a positive displacement method, meaning that

material flow is generated directly by piston movement. As the piston advances, a corresponding volume of material is displaced and forced through the nozzle.

This approach provides several advantages for frosting applications:

- High force generation capability,
- Controlled volumetric deposition,
- Consistent flow characteristics,
- Suitability for highly viscous materials,
- Straightforward flow-rate control,
- Relatively simple mechanical implementation.

Because material displacement is directly related to piston movement, syringe systems can provide improved deposition repeatability compared to gravity-fed systems. The ability to generate substantial extrusion forces also makes them suitable for frosting materials exhibiting higher yield stresses and viscosities.

Additionally, syringe systems can be integrated effectively with CNC-based motion systems, enabling synchronized control between frosting deposition and nozzle positioning.

The primary limitations of syringe systems include finite material capacity and the need for periodic refilling[19]. However, these limitations are generally acceptable in prototype and small-scale automated frosting applications where deposition consistency is prioritized over continuous industrial production.

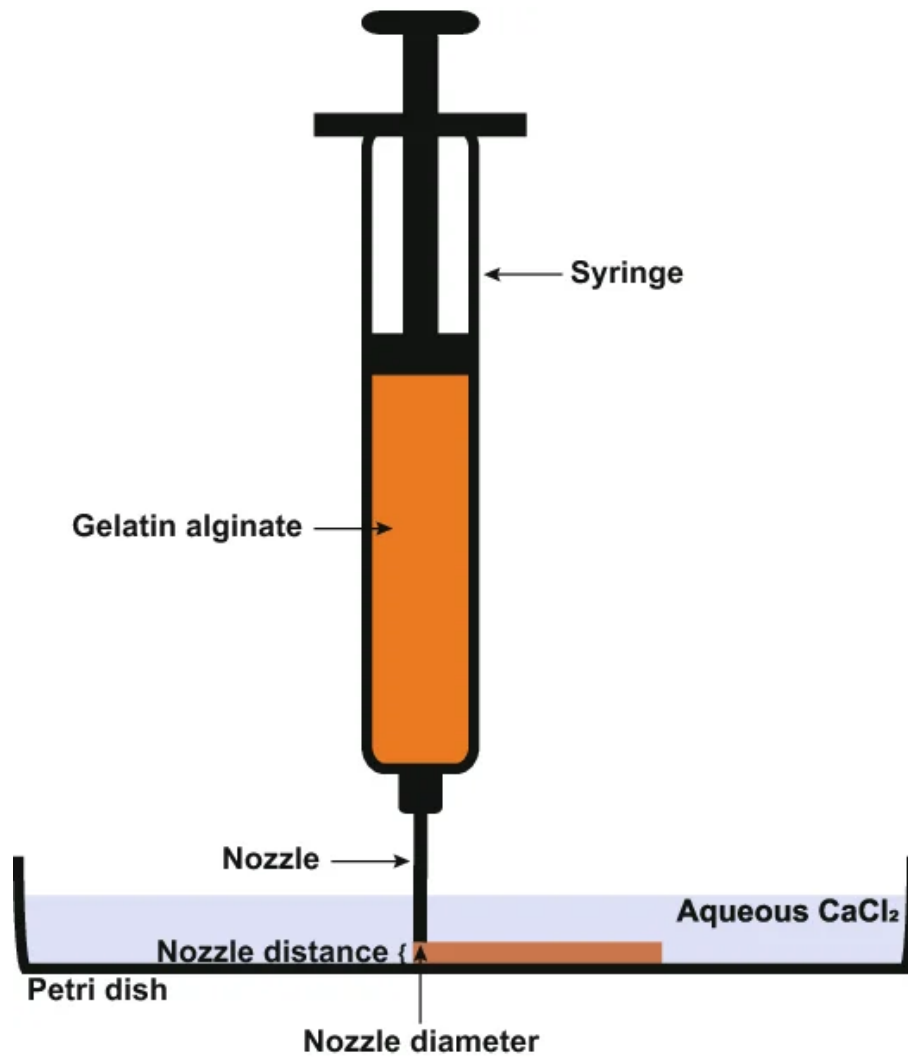


Figure 2.10: Syringe-based extrusion system

2.4.5 Comparison of Extrusion Systems for Frosting Applications

The suitability of an extrusion system for cake frosting depends on its ability to generate adequate pressure, maintain consistent flow, and integrate effectively with the machine's motion system.

Gravity-fed systems offer simplicity but provide limited force generation and poor flow consistency for highly viscous materials. Auger systems are capable of continuous material delivery but may introduce flow variations and increased mechanical complexity. Pneumatic systems provide smooth operation but require additional infrastructure and may present challenges in precise volumetric control.

Syringe-based systems provide a balance between force generation, deposition accuracy, and mechanical simplicity. By directly linking piston displacement to material flow, these systems offer improved control over frosting deposition while remaining compatible with automated motion control architectures.

Consequently, syringe-based extrusion systems are widely adopted in research-scale food deposition equipment, food printing systems, and automated frosting applications involving medium- to high-viscosity materials.

2.5 Motion Systems in Automated Manufacturing

Motion systems form the foundation of automated manufacturing equipment by providing controlled movement of tools, workpieces, or both within a defined workspace. The selection of an appropriate motion system significantly influences machine performance, positional accuracy, structural rigidity, control complexity, manufacturing cost, and ease of implementation.

In automated cake frosting applications, the motion system determines how accurately the frosting nozzle can be positioned relative to the cake surface. Since frosting quality depends on consistent nozzle placement and synchronized material deposition, the choice of motion architecture is a critical design consideration.

Several motion system configurations are commonly employed in automated manufacturing and robotic applications, including Cartesian, SCARA, and Delta systems. Each offers unique advantages and limitations depending on the required workspace, speed, payload capacity, and control requirements[4,13].

2.5.1 Cartesian Motion Systems

Cartesian motion systems utilize independent linear movement along mutually perpendicular axes, typically designated as X, Y, and Z. Positioning is achieved by controlling movement along each axis individually, allowing the tool position to be defined directly using Cartesian coordinates.

Because movement occurs along orthogonal axes, Cartesian systems possess relatively simple kinematic relationships compared to many robotic alternatives. This simplifies motion planning, machine calibration, and software implementation.

Cartesian systems are widely used in:

- CNC machines,
- 3D printers,
- automated dispensing systems,
- pick-and-place equipment,
- laboratory automation systems.

Major advantages include:

- straightforward control,
- predictable workspace geometry,
- high positioning accuracy,
- ease of calibration,
- good structural rigidity.

However, Cartesian systems may require larger structural frameworks than some robotic configurations and can experience increased moving mass depending on axis arrangement.

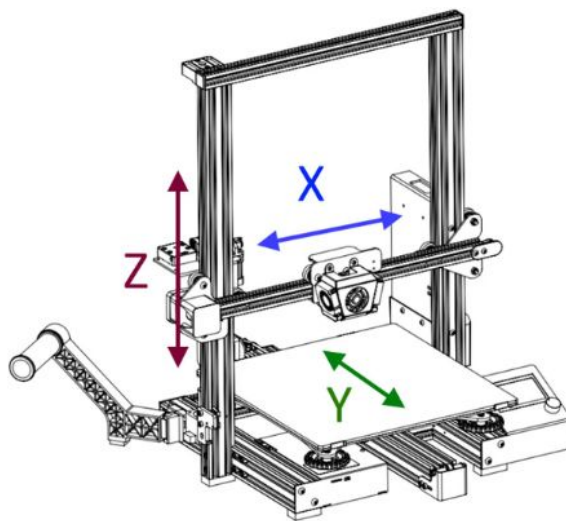


Figure 2.11: Typical cartesian motion system

2.5.2 SCARA Motion Systems

Selective Compliance Assembly Robot Arm (SCARA) systems employ a serial robotic architecture consisting of rotational joints operating primarily within a horizontal plane. SCARA robots are widely used in assembly, packaging, material handling, and high-speed pick-and-place operations[13].

Their articulated design enables rapid movement and efficient operation within relatively compact workspaces.

Advantages of SCARA systems include:

- High operating speeds,
- Compact machine footprint,
- Efficient use of workspace,
- Good repeatability.

However, SCARA systems require more complex kinematic calculations than Cartesian systems. The articulated structure also introduces additional control complexity and may complicate the generation of frosting paths on cylindrical cake surfaces.

While SCARA robots are effective for many industrial automation tasks, their advantages are not fully utilized in applications requiring relatively simple and predictable frosting trajectories.

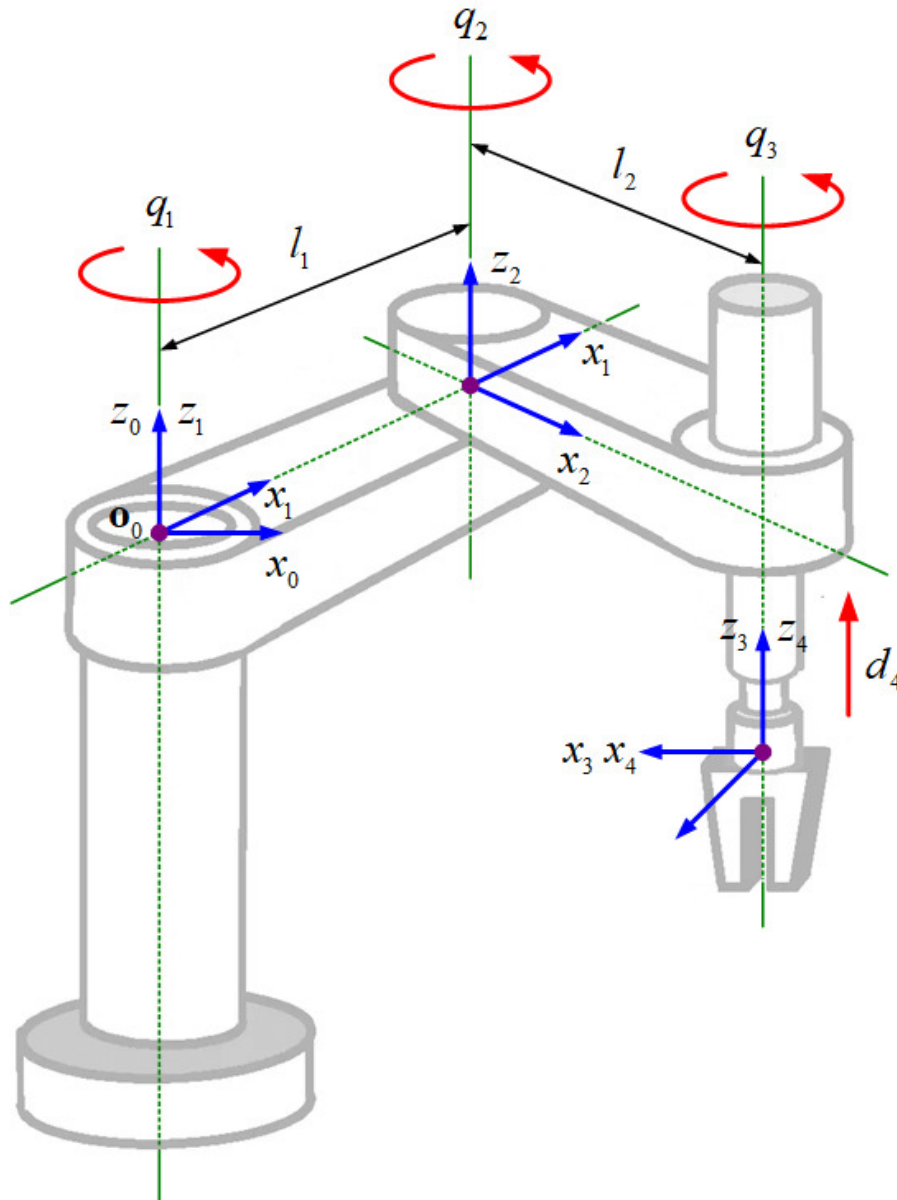


Figure 2.12: SCARA system

2.5.3 Delta Motion Systems

Delta robots are parallel robotic systems consisting of multiple lightweight arms connected to a common end effector. The parallel architecture enables high-speed movement and low moving mass, making Delta systems particularly attractive for food packaging and sorting applications.

Delta robots are commonly used in:

- food handling systems,

- packaging operations,
- pharmaceutical manufacturing,
- high-speed sorting applications.

Advantages include:

- very high operating speeds,
- low moving inertia,
- excellent dynamic performance.

Despite these benefits, Delta systems possess more complex kinematic relationships and generally operate within non-rectangular workspaces. The mechanical and software complexity associated with Delta robots can increase design and implementation requirements.

For applications focused primarily on controlled frosting deposition rather than extremely high-speed operation, the additional complexity may not provide significant practical benefits.

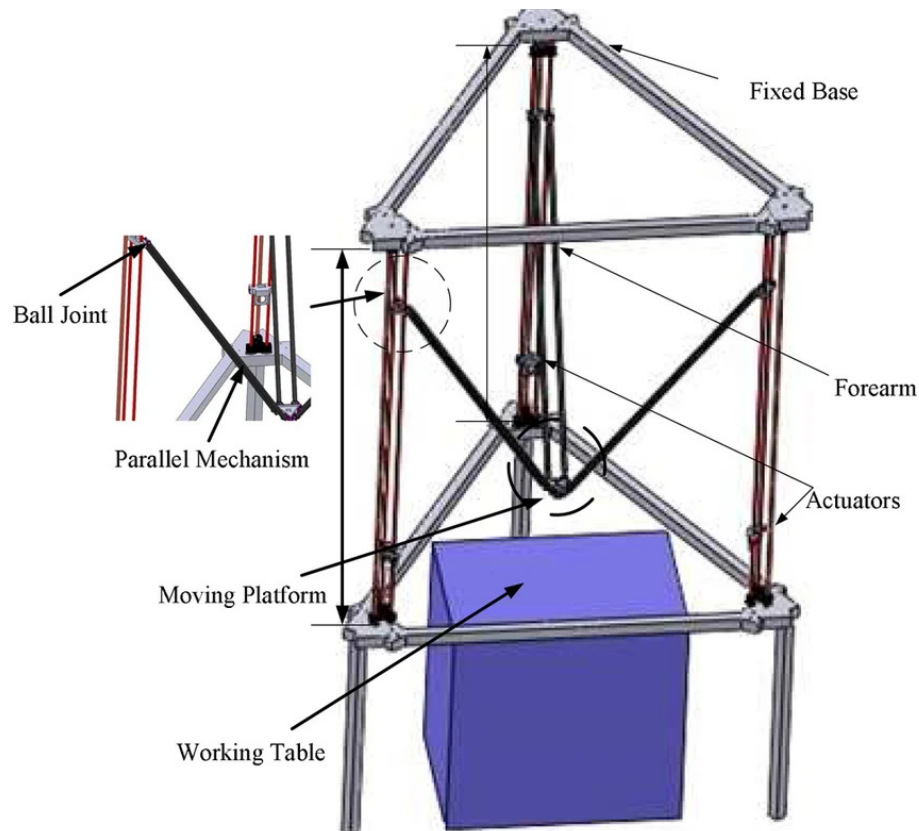


Figure 2.13: Delta motion system

2.5.4 Gantry-Based Cartesian Systems

Gantry systems represent a specialized form of Cartesian motion architecture in which one axis is supported by a bridge structure spanning the machine workspace. The gantry carries additional motion axes and enables positioning over a large working area while maintaining the simplicity of Cartesian motion control.

Because gantry systems retain Cartesian kinematics, they combine straightforward control with excellent workspace accessibility. Two common gantry configurations are fixed gantry systems and moving gantry systems.

Fixed Gantry Systems

In a fixed gantry configuration, the bridge structure remains stationary while the workpiece or lower platform moves beneath it.

Advantages include:

- High structural rigidity,
- Reduced moving mass,
- Good positional accuracy.

However, the moving workpiece arrangement may increase machine complexity and can be less suitable for applications involving larger or heavier products.



Figure 2.14: Fixed gantry system

Moving Gantry Systems

In a moving gantry configuration, the gantry bridge itself translates across the machine frame while carrying one or more additional motion axes.

Advantages include:

- efficient workspace utilization,
- simplified machine layout,
- improved accessibility,
- straightforward integration of additional axes,
- suitability for medium-sized workspaces.

Moving gantry systems are widely employed in:

- CNC routers,
- laser cutting systems,
- additive manufacturing equipment,
- automated dispensing systems.

For automated cake frosting applications, the moving gantry arrangement allows the frosting nozzle to access both the top surface and side regions of the cake while maintaining an unobstructed workspace around the rotary platform. The configuration also simplifies integration with extrusion systems and auxiliary tooling.



Figure 2.15: Moving gantry system

2.6 Rotary Positioning Systems

Rotary positioning systems are widely used in manufacturing and automation to provide controlled angular movement of a workpiece, tool, or assembly. Unlike linear motion systems that move along straight paths[12], rotary systems generate rotational motion about a fixed axis. Such systems are commonly employed when a process requires continuous circumferential access to a component or when rotational positioning offers a simpler alternative to complex multi-axis linear movement.

In automated cake frosting applications, rotational motion provides an efficient method of exposing the entire circumference of the cake to the frosting nozzle. Rather than moving the nozzle completely around the cake, the cake itself may be rotated beneath a stationary or partially moving nozzle. This approach simplifies motion planning and can improve frosting consistency on cylindrical surfaces.

2.6.1 Rotary Tables in Manufacturing

Rotary tables are positioning devices designed to rotate a workpiece through controlled angular increments or continuous rotational motion. They are extensively used in manufacturing environments where multiple surfaces of a component must be accessed without repeatedly repositioning the workpiece.

Common applications include:

- CNC machining operations,
- automated assembly systems,
- indexing mechanisms,
- inspection systems,
- packaging equipment.

Rotary tables improve process efficiency by reducing the number of linear movements required to access different regions of a workpiece. In many CNC applications, a rotary table effectively functions as an additional axis of motion[15], expanding the machine's operational capabilities.

The precision of a rotary table depends on factors such as drive mechanism selection, bearing support, structural rigidity, and motor control accuracy.



Figure 2.16: Industrial rotary table

2.6.2 Direct Drive Rotary Systems

Direct drive rotary systems connect the motor directly to the rotating platform without intermediate transmission elements. The motor shaft itself forms the rotational axis, eliminating belts, gears, and other mechanical transmission components.

Advantages include:

- High positional accuracy,
- Minimal backlash,
- Smooth motion characteristics,
- Reduced mechanical complexity.

Because fewer mechanical components are present, direct drive systems can achieve excellent positioning performance and reduced maintenance requirements.

However, direct drive arrangements often require larger motors to generate sufficient torque, particularly when rotating heavier loads. They may also increase system cost due to the precision components involved.

For small rotational loads, direct drive systems can provide an effective and compact solution. However, larger applications may require additional torque multiplication through alternative transmission methods.



Figure 2.17: Direct rotary drive

2.6.3 Belt Driven Rotary Systems

Belt driven rotary systems utilize timing belts and pulleys to transmit rotational motion from a motor to the rotating platform. Timing belts provide positive engagement between pulleys, ensuring synchronized motion without slippage under normal operating conditions.

Advantages include:

- Mechanical simplicity,
- Relatively low cost,
- Ease of implementation,
- Flexible component placement,
- Vibration isolation between motor and platform.

An additional benefit of belt-driven systems is the ability to modify torque and rotational speed through pulley size selection. By employing reduction ratios, greater rotational torque can be achieved without increasing motor size.

Potential limitations include:

- Belt wear over time,
- Tension adjustment requirements,
- Reduced stiffness compared to direct drive systems.

Despite these limitations, belt-driven rotary mechanisms are widely used in automation equipment due to their practicality and ease of fabrication.



Figure 2.18: Belt-driven rotary system

2.6.4 Gear Driven Rotary Systems

Gear driven rotary systems utilize meshing gears to transfer rotational motion between the motor and rotating platform. Spur gears, worm gears, planetary gear systems, and

ring gears are commonly employed depending on application requirements[6,7].

Advantages include:

- High torque transmission capability,
- Compact packaging,
- Precise speed reduction,
- Suitability for heavy loads.

Gear systems are particularly useful where substantial torque multiplication is required or where rotational motion must be maintained under significant loading conditions.

However, gear-driven systems may introduce backlash, increase manufacturing complexity, and require more precise alignment during assembly. Lubrication and maintenance requirements may also be greater than those of belt-driven alternatives.

While gear systems are common in industrial positioning equipment, their additional complexity may not always be justified in smaller automated deposition systems.

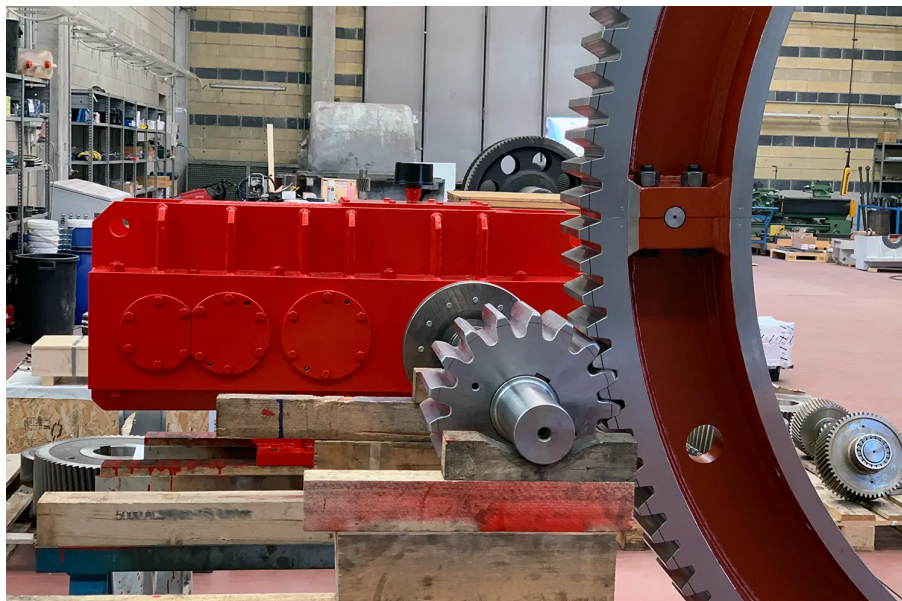


Figure 2.19: Gear-driven rotary system

2.6.5 Rotary Systems in Food Processing

Rotary positioning systems are widely utilized throughout the food processing industry. Their ability to provide controlled and repeatable rotational motion makes them suitable

for numerous food handling and processing operations[10].

Examples include:

- Cake decorating turntables,
- Automated icing and frosting systems,
- Food inspection platforms,
- Packaging conveyors,
- Rotary filling systems,
- Product handling equipment.

In cake decoration, rotating platforms have long been used manually to assist decorators in applying frosting uniformly around the circumference of a cake. The integration of motorized rotary systems extends this concept by providing controlled rotational speed and repeatable positioning.

For automated frosting applications, rotational motion offers several advantages. By rotating the cake beneath the frosting nozzle, the machine can maintain a relatively simple nozzle trajectory while achieving continuous frosting deposition around cylindrical surfaces. This reduces the complexity of the motion system and improves the uniformity of side-wall frosting operations.

Additionally, rotary motion can be synchronized with extrusion rate and nozzle positioning to improve frosting consistency and surface quality[1].



Figure 2.20: Rotary table for cake frosting

2.7 Motion Transmission Systems

Motion transmission systems are responsible for converting and transferring mechanical power between actuators and machine components. In automated systems, the selection of an appropriate transmission mechanism influences positioning accuracy, force generation capability, motion smoothness, synchronization, and overall machine performance.

Automated cake frosting machines require both accurate positional control and controlled material extrusion. Consequently, different transmission technologies may be employed depending on the specific function being performed. Linear positioning systems typically prioritize accuracy and repeatability, whereas extrusion systems require sufficient force generation to overcome frosting resistance during deposition.

This section reviews the motion transmission components relevant to automated frosting applications.

2.7.1 DC Motors

Direct current (DC) motors convert electrical energy into continuous rotational motion. Unlike stepper motors, which move in discrete increments, DC motors rotate continuously when supplied with electrical power and are commonly controlled by varying the applied voltage or pulse-width modulation (PWM) signal. DC motors are widely used in:

- pumps,
- conveyors,
- dispensing systems,
- mixers,
- extrusion mechanisms.

Their relatively simple control requirements make them suitable for applications where precise positional control is less important than continuous and adjustable rotational motion.

In automated frosting systems, DC motors can be used to drive extrusion mechanisms where frosting flow rate is regulated by adjusting motor speed. This allows material delivery to be varied dynamically during operation to match changing deposition requirements.

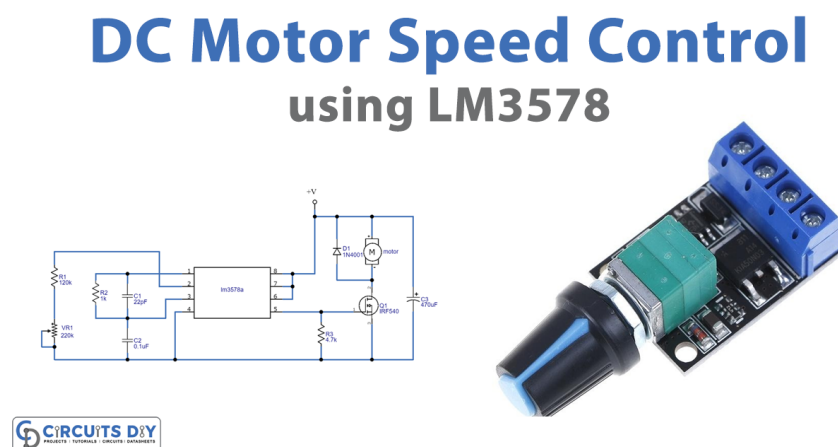


Figure 2.21: DC motor setup

2.7.2 Stepper Motors

Stepper motors are electromechanical actuators that divide rotational motion into discrete angular increments known as steps. Each electrical pulse supplied to the motor produces a predictable angular displacement, allowing position to be controlled through pulse counting.

Because of their ability to provide accurate open-loop positioning, stepper motors are widely used in:

- CNC machines,
- 3D printers,
- robotic systems,
- automated dispensing equipment,
- laboratory automation systems.

The principal advantages of stepper motors include:

- precise positional control,
- repeatable movement,
- ease of digital control,
- compatibility with CNC motion systems.

Through microstepping techniques, motion smoothness and positioning resolution can be further improved.

For automated cake frosting applications, stepper motors are particularly well suited for controlling the X, Y, Z, and rotary axes where accurate nozzle positioning and coordinated motion are required.

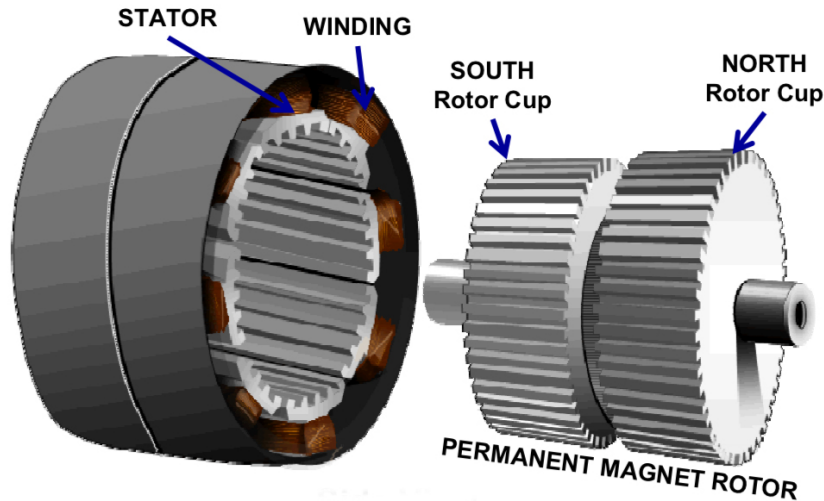


Figure 2.22: Stepper motor

2.7.3 Lead Screw Mechanisms

Lead screw mechanisms convert rotational motion into linear displacement through the interaction of a threaded screw and mating nut. As the screw rotates, the nut advances along the screw axis, producing controlled linear movement.

One of the most important characteristics of lead screw systems is their ability to convert relatively small motor torques into significant linear forces. This mechanical advantage makes lead screws particularly useful in applications involving lifting, pressing, or extrusion.

Lead screw systems are commonly used in:

- CNC positioning systems,
- vertical motion axes,
- syringe extrusion mechanisms,
- automated dispensing equipment.

Advantages include:

- high force generation capability,
- accurate linear positioning,

- controlled motion,
- suitability for high-load applications.

In frosting deposition systems, lead screws are especially attractive for syringe-based extrusion because they enable the generation of sufficient force to overcome frosting yield stress and maintain consistent material flow through the nozzle[8].



Figure 2.23: Lead screw mechanism

2.7.4 Timing Belt Systems

Timing belts are positive-drive transmission elements that transfer motion between pulleys through the engagement of matching teeth. Unlike conventional belts, timing belts maintain a fixed positional relationship between driving and driven components, preventing slippage during operation.

Timing belt systems are widely used in:

- CNC machines,

- gantry systems,
- 3D printers,
- conveyor systems,
- automated manufacturing equipment.

Their popularity results from several advantages:

- lightweight construction,
- low cost,
- low maintenance requirements,
- smooth operation,
- ability to transmit motion over relatively long distances.

Timing belts are also frequently employed to synchronize multiple shafts or motion components. In dual-drive systems, a timing belt can ensure that both sides of a mechanism move together, reducing skewing and improving positional consistency.

For automated frosting machines, timing belts may be used to synchronize moving components while maintaining relatively simple mechanical construction.



Figure 2.24: Timing belt and pulley

2.7.5 Shaft Couplers

Mechanical systems often require the connection of rotating shafts possessing different diameters, manufacturing tolerances, or alignment conditions. Shaft couplers are therefore used to transmit torque between a driving shaft and a driven shaft while maintaining reliable power transfer.

Several coupler designs are commonly employed in automation systems.

Rigid Couplers

Rigid couplers provide a direct mechanical connection between shafts and offer high torsional stiffness. They are simple and effective when accurate shaft alignment can be maintained.

Advantages include:

- efficient torque transmission,
- high stiffness,
- simple construction.

However, rigid couplers are sensitive to misalignment and may transmit vibration directly between connected components.



Figure 2.25: Rigid coupler

Flexible Couplers

Flexible couplers incorporate features that allow limited compensation for angular, axial, or parallel misalignment.

Advantages include:

- reduced vibration transmission,

- accommodation of alignment errors,
- reduced mechanical stress.

These characteristics make flexible couplers common in CNC and automation equipment where perfect alignment is difficult to achieve.



Figure 2.26: Flexible couplers

Custom-Manufactured Couplers

In prototype and research systems, commercially available couplers may not always match the required shaft dimensions or mounting configurations. Under such circumstances, custom-manufactured couplers can be produced through machining, turning, additive manufacturing, or a combination of fabrication methods.

Custom couplers provide:

- dimensional flexibility,
- rapid implementation,
- integration with application-specific requirements.

They are particularly useful when connecting shafts of non-standard diameters or when adapting readily available components into a custom machine design.

For automated frosting equipment, custom couplers may provide a practical solution for connecting motor shafts to lead screws while maintaining reliable torque transmission and simplifying system integration.

2.8 Embedded CNC Control Systems

Modern automated manufacturing systems rely on embedded control technologies to coordinate machine motion, execute programmed tasks, monitor operating conditions, and provide interfaces through which operators can interact with the equipment. Advances in microcontrollers, wireless communication technologies, and software development platforms have enabled increasingly sophisticated automation systems to be implemented using compact and cost-effective hardware.

In automated cake frosting applications, the control system serves as the central component responsible for coordinating motion, regulating frosting deposition, executing programmed frosting routines, and managing user interaction. The effectiveness of the machine therefore depends not only on the mechanical design but also on the capability of the embedded control architecture[20].

2.8.1 CNC Motion Controllers

Computer Numerical Control (CNC) systems are widely used in manufacturing to automate machine movement through programmed instructions. CNC controllers interpret motion commands and generate coordinated signals that drive motors along predefined paths.

Traditional CNC systems are commonly used in:

- milling machines,
- lathes,
- laser cutters,
- plasma cutting systems,
- additive manufacturing equipment.

A typical CNC controller is responsible for:

- motion planning,
- position control,

- trajectory generation,
- multi-axis synchronization,
- process execution.

By coordinating multiple axes simultaneously, CNC systems enable complex tool paths to be executed with high repeatability and accuracy. This capability has led to their adoption in numerous automated deposition applications where controlled positioning is required.

In frosting applications, CNC principles can be adapted to control nozzle movement, cake positioning, and material deposition in a coordinated manner.

2.8.2 ESP32-Based Machine Control

Recent developments in embedded electronics have enabled compact microcontrollers to perform many functions traditionally associated with larger industrial control systems. One such platform is the ESP32 family of microcontrollers, which combines processing capability, wireless communication, and flexible input-output functionality within a single device.

The ESP32 platform offers several advantages for automation applications:

- integrated Wi-Fi connectivity,
- high processing capability,
- multiple communication interfaces,
- pulse generation capability for motor control,
- low power consumption,
- relatively low implementation cost.

These features have contributed to the widespread adoption of ESP32-based systems in robotics, CNC control, Internet of Things (IoT) devices, and industrial automation research.

For automated frosting machines, an ESP32-based controller can coordinate motor motion, regulate extrusion systems, execute frosting algorithms, and provide wireless

communication with user interfaces. The integration of motion control and networking capabilities within a single platform reduces hardware complexity while maintaining system flexibility.

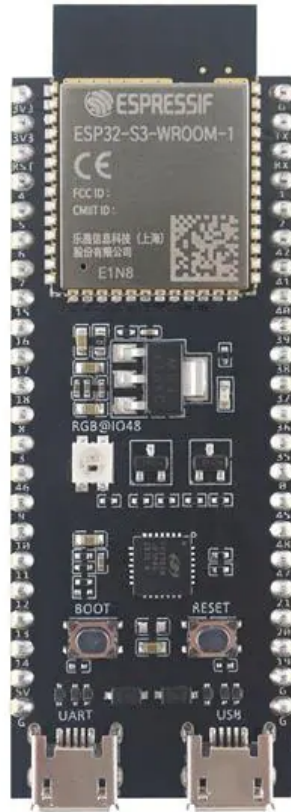


Figure 2.27: ESP32

2.8.3 Web-Enabled Machine Control

The incorporation of wireless communication technologies has transformed the way operators interact with automated equipment. Rather than relying exclusively on dedicated control panels, modern systems can provide browser-based interfaces accessible through smartphones, tablets, and personal computers.

Web-enabled control systems typically consist of:

- an embedded web server,
- wireless communication infrastructure,
- browser-based user interfaces,

- real-time machine monitoring functions.

Such systems offer several advantages:

- remote accessibility,
- reduced hardware requirements,
- simplified user interaction,
- platform independence,
- real-time parameter adjustment.

In automation applications, wireless dashboards enable operators to monitor machine status, modify operating parameters, initiate automated routines, and respond to system events without direct physical interaction with the controller hardware.

For research and prototype systems, web-based interfaces provide a flexible and cost-effective alternative to dedicated industrial control panels.

2.8.4 Human-Machine Interfaces

A Human-Machine Interface (HMI) serves as the communication link between the operator and an automated system. HMIs allow users to monitor machine conditions, issue commands, adjust process parameters, and receive operational feedback.

Effective HMIs contribute significantly to system usability by simplifying interaction with complex machine functions.

Common HMI functions include:

- machine control,
- parameter configuration,
- status monitoring,
- fault reporting,
- process visualization.

The design of an HMI should prioritize clarity, ease of use, and accessibility. Information such as machine position, operating mode, system status, and process variables should be presented in a manner that enables rapid interpretation by the operator.

In automated frosting systems, the HMI may be used to:

- configure frosting parameters,
- control motion systems,
- initiate frosting routines,
- monitor machine status,
- perform calibration procedures.

A well-designed interface improves operational efficiency while reducing the likelihood of user error during machine operation.

2.8.5 Automated Food Deposition Control

Automated food deposition systems require more than simple motion control. The deposition process must coordinate material flow with machine movement to ensure that the desired quantity of material is applied at the correct location and at the appropriate rate.

Three key aspects of automated deposition control are synchronization, flow control, and trajectory execution.

Synchronization

Synchronization refers to the coordination of multiple machine functions during operation. In frosting applications, nozzle motion, cake rotation, and extrusion activity must operate together to produce consistent frosting coverage.

Poor synchronization can result in:

- frosting overlap,
- uneven deposition,
- gaps in coverage,
- inconsistent surface finish.

Effective synchronization therefore plays a critical role in maintaining frosting quality.

Flow Control

Flow control governs the rate at which frosting is delivered through the extrusion system. Because frosting materials may exhibit varying viscosities and yield stresses, the flow rate must often be adjusted to match machine motion.

Accurate flow control contributes to:

- uniform frosting thickness,
- improved deposition consistency,
- reduced material waste,
- enhanced surface quality.

Maintaining a proper balance between extrusion rate and nozzle velocity is particularly important during automated frosting operations.

Trajectory Execution

Trajectory execution involves moving the deposition nozzle along predefined paths while maintaining appropriate process conditions. Depending on cake geometry, trajectories may consist of linear paths, circular motions, spirals, raster patterns, or combinations of multiple motion types.

The ability to execute repeatable trajectories enables automated systems to produce consistent frosting patterns and decorative features while reducing dependence on operator skill.

In advanced frosting systems, trajectory execution may be combined with adaptive flow control and synchronized rotational motion to improve frosting uniformity across both horizontal and vertical cake surfaces.

2.9 Challenges in Automated Food Deposition Systems

Despite significant advances in automation and food processing technologies, automated deposition of semi-solid food materials remains a challenging engineering problem. The

combination of complex material behaviour, mechanical system limitations, and process variability can negatively affect product quality and system reliability.

In cake frosting applications, successful operation depends on the ability to consistently deliver frosting through an extrusion system while maintaining accurate positioning and uniform deposition. Various studies and industrial implementations have identified several recurring challenges associated with automated food deposition systems.

2.9.1 Nozzle Clogging

Nozzle clogging is one of the most common challenges encountered in food extrusion and deposition systems. Clogging occurs when material accumulates within the nozzle or when the flow passage becomes partially obstructed.

Several factors contribute to nozzle clogging, including:

- high material viscosity,
- inadequate extrusion pressure,
- small nozzle diameters,
- material drying at the nozzle outlet,
- the presence of lumps or unmixed ingredients.

Clogging can interrupt material flow, produce inconsistent deposition patterns, and increase process downtime. In frosting applications, nozzle blockage may result in uneven frosting coverage and poor surface finish[2].

The selection of appropriate nozzle geometry, extrusion force, and frosting preparation procedures is therefore important in reducing the likelihood of clogging during operation.

2.9.2 Frosting Viscosity Variation

The rheological properties of frosting materials are influenced by numerous factors including formulation, ingredient ratios, mixing procedures, temperature, and storage conditions. As a result, frosting viscosity may vary significantly between production batches or even during a single operating cycle.

Variations in viscosity can affect:

- extrusion force requirements,
- material flow rate,
- frosting thickness,
- surface quality,
- deposition repeatability.

A frosting formulation that flows adequately under one set of conditions may require substantially greater extrusion force under different environmental or preparation conditions.

This variability presents a challenge for automated systems because machine settings that perform well for one frosting condition may not produce the same results under another. Consequently, frosting deposition systems must possess sufficient force capability and adjustment flexibility to accommodate expected material variations.

2.9.3 Extrusion Control Challenges

Maintaining a consistent extrusion rate is essential for achieving uniform frosting deposition. However, several factors can influence material delivery during operation.

These include:

- changes in frosting viscosity,
- pressure fluctuations,
- nozzle restrictions,
- mechanical backlash,
- variations in extrusion speed.

Inconsistent extrusion may lead to:

- frosting buildup,
- under-deposition,
- irregular surface finishes,

- decorative defects.

For highly viscous frosting materials, insufficient force generation can prevent reliable material flow. Conversely, excessive extrusion may produce undesirable frosting accumulation and reduced dimensional accuracy.

Effective extrusion control therefore requires careful matching of force generation capability, nozzle geometry, and deposition speed.

2.9.4 Motion Synchronization

Automated frosting operations often require multiple machine functions to operate simultaneously. These functions may include nozzle positioning, rotational motion, vertical movement, and frosting extrusion.

Synchronization errors occur when these subsystems fail to operate in a coordinated manner.

Common consequences include:

- frosting overlap,
- gaps in frosting coverage,
- inconsistent frosting thickness,
- distorted decorative patterns,
- reduced repeatability.

The challenge becomes particularly significant when frosting cylindrical cake surfaces where rotational motion must be coordinated with nozzle position and material flow rate.

Accurate synchronization is therefore essential for maintaining frosting quality and achieving consistent results across multiple production cycles.

2.9.5 Structural Vibration and Mechanical Stability

Mechanical vibration and structural deflection can negatively affect positioning accuracy during automated deposition processes. Sources of vibration may include motor operation, rapid acceleration, transmission mechanisms, and insufficient frame rigidity.

In frosting applications, even small positional deviations can influence frosting thickness and surface appearance.

Potential consequences include:

- uneven frosting layers,
- surface imperfections,
- reduced dimensional accuracy,
- inconsistent decorative features.

Machine structures must therefore provide sufficient rigidity to support motion components while minimizing unwanted movement during operation. Proper mechanical design contributes significantly to process repeatability and overall machine performance.

2.9.6 Cleaning and Food Safety Requirements

Food processing equipment must be designed with consideration for sanitation, hygiene, and ease of cleaning. Frosting materials contain ingredients that may support microbial growth if residues are allowed to accumulate within the machine.

Areas of concern include:

- material reservoirs,
- nozzles,
- tubing,
- extrusion chambers,
- food-contact surfaces.

Complex geometries and inaccessible components can increase cleaning difficulty and extend maintenance time.

In addition to maintaining product quality, proper cleaning practices are necessary to comply with food safety requirements and prevent contamination between production batches. Consequently, ease of cleaning is an important consideration in the design of automated frosting systems.

Chapter 3

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter presents the methodology adopted for the design, fabrication, integration, and testing of the automated cake frosting machine. A requirement-driven engineering approach was employed, where design decisions were based on the selected frosting material and the operational requirements of automated cake frosting.

The methodology progressed from frosting material selection and requirement identification to subsystem design, fabrication, software implementation, and system integration.

3.2 Design Methodology

The development process followed a structured engineering workflow consisting of:

- Requirement identification,
- Concept development,
- Subsystem design,
- CAD modelling,
- Fabrication,
- Integration,
- Testing and evaluation.

Literature Review → Requirement Identification → Concept Development → Design
→ Fabrication → Integration → Testing → Evaluation

The design process began with the selection of the frosting material since frosting characteristics directly influence extrusion, nozzle design, motion control, and machine operation.

3.3 Frosting Material Selection and Engineering Requirements

3.3.1 Evaluation of Frosting Materials

Several frosting materials were considered including:

- Whipped cream,
- Buttercream frosting,
- Cream cheese frosting,
- Royal icing.

The evaluation considered:

- Extrusion behaviour,
- Shape retention,
- Stability after deposition,
- Suitability for automated frosting operations.

3.3.2 Selection of Buttercream Frosting

Buttercream frosting was selected due to its:

- Widespread use in cake decoration,
- Good structural stability,
- Suitability for piping and smoothing operations,
- Relevance to commercial cake production.

As buttercream requires greater extrusion force than lower-viscosity materials, it provided a suitable basis for machine design.

3.3.3 Rheological Characteristics

Buttercream exhibits shear-dependent flow behaviour, allowing it to flow during extrusion while maintaining its shape after deposition.

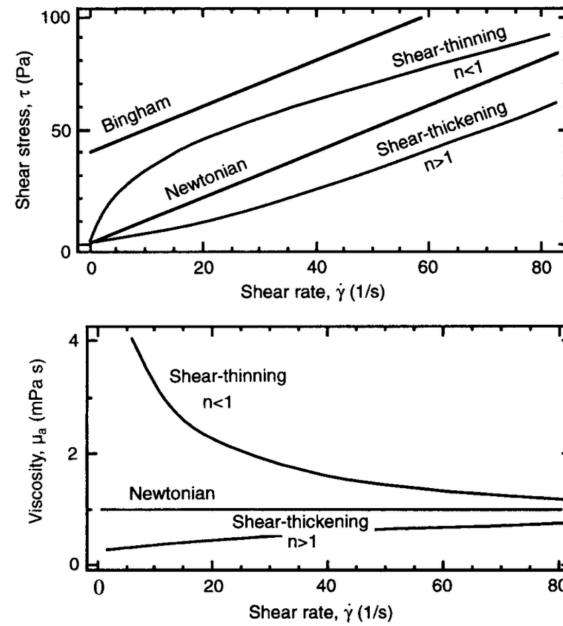


Figure 3.1: Comparison between generic frosting, and shear- thinning frosting materials

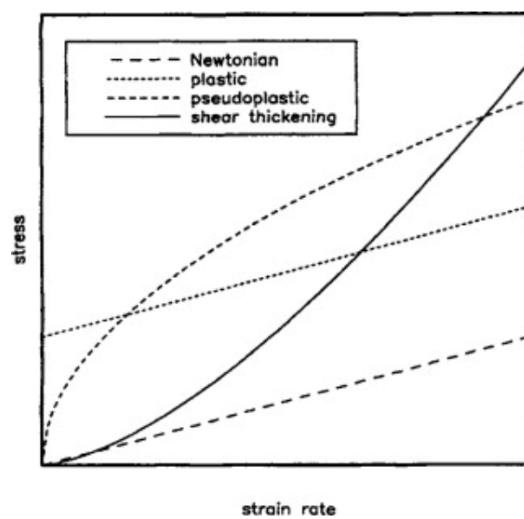


Figure 3.2: Idealized Rheological Behaviour of Frosting Materials

The rheological characteristics of buttercream indicate the need for controlled force generation and consistent material delivery.

3.3.4 Derived Engineering Requirements

The selected frosting material resulted in the following design requirements:

- Requirement 1: Generate sufficient extrusion force.
- Requirement 2: Maintain consistent frosting flow.
- Requirement 3: Support top and side-wall frosting.
- Requirement 4: Provide synchronized motion control.
- Requirement 5: Allow programmable frosting patterns.

These requirements formed the basis for the design of the machine subsystems presented in the following sections.

3.4 Overall Machine Concept

Based on the engineering requirements established in Section 3.3, an automated cake frosting machine was developed to provide controlled frosting deposition on both the top and side surfaces of cakes.

The proposed machine consists of five primary subsystems:

- Motion system
- Structural frame
- Theta-axis rotary platform
- Frosting extrusion system
- Control and electrical system

A moving-gantry Cartesian architecture was selected to position the frosting nozzle above the cake, while a motorized rotary platform was incorporated to facilitate continuous side-wall frosting operations. Frosting delivery is achieved through a lead screw-driven syringe extrusion mechanism capable of generating sufficient force for buttercream deposition.

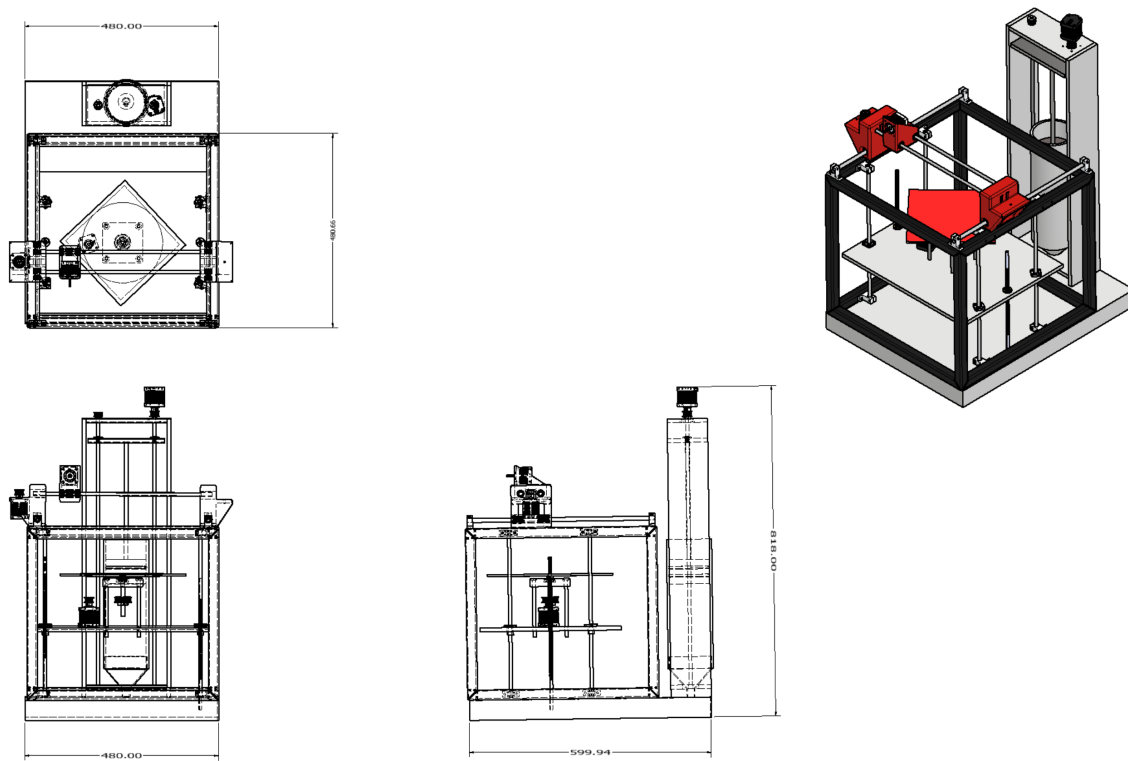


Figure 3.3: The Overall Machine Architecture

Subsystem labels:

X-axis

Y-axis

Z-axis

Rotary platform

Extrusion assembly

Control enclosure

Frosting nozzle

The interaction between the major subsystems is illustrated in Figure 3.4.

The selected architecture satisfies the engineering requirements for controlled extrusion, synchronized motion, side-wall frosting capability, and programmable operation.

3.5 Motion System Selection and Design

3.5.1 Motion System Evaluation

Three motion system architectures were considered during concept development:

Because frosting operations require accurate positioning rather than extremely high operating speeds, the Cartesian configuration was selected.

3.5.2 Moving Gantry Configuration

Following selection of the Cartesian architecture, a moving gantry configuration was adopted.

Advantages include:

- Unobstructed work area
- Simple machine layout
- Accessibility to cake surface
- Ease of integration with extrusion system
- Compatibility with rotary platform operation

The moving gantry supports the X, Y, and Z motion axes while maintaining adequate access to the frosting workspace.

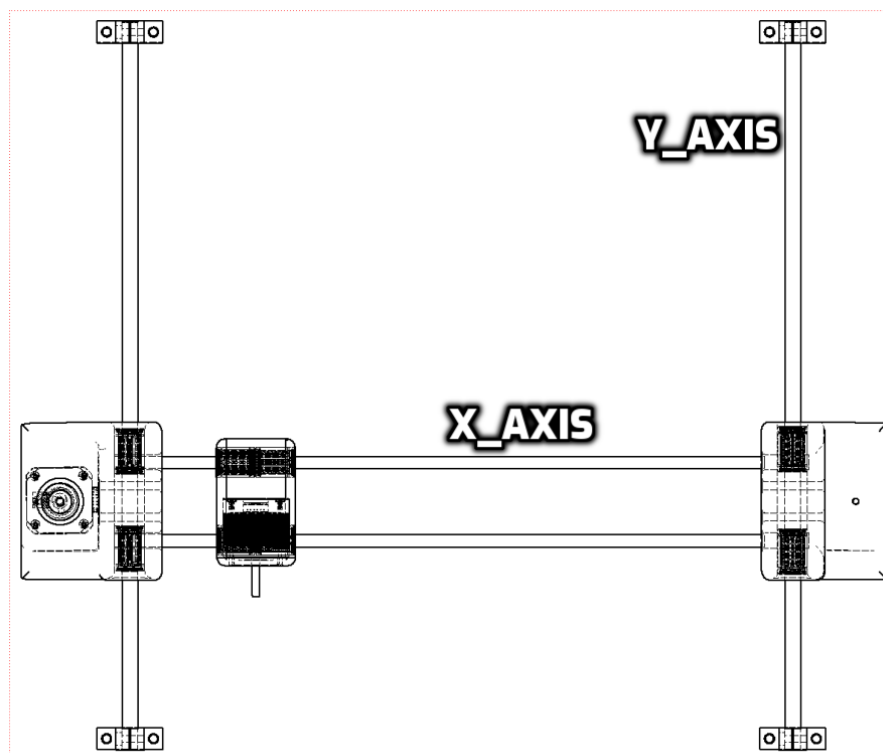


Figure 3.4: shows the X and Y moving gantry configuration.

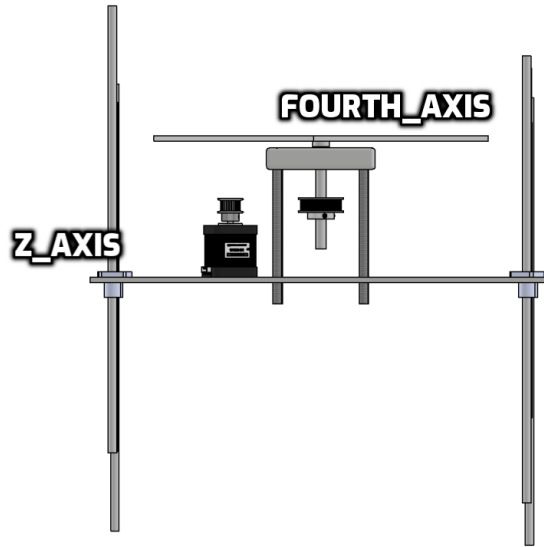


Figure 3.5: Z and 4TH Axis

3.5.3 Axis Travel Design

The machine workspace was determined from the maximum anticipated cake dimensions and required tooling clearances.

Travel requirements were established as:

Additional travel allowances were incorporated to accommodate:

- Nozzle clearance
- Calibration procedures
- Tool changes
- Frosting operations

3.5.4 CAD Model

The complete motion system was modelled using CAD software prior to fabrication.

The CAD model was used to:

- Verify workspace accessibility
- Evaluate component clearances

-
- This diagram illustrates the assembly of the top carriage and top rail. A red carriage (3) is being mounted onto the top rail (4), which is being inserted into the top of the frame. The carriage is shown in the process of being slid onto the rail, with a red arrow indicating the direction of movement. The rail is being inserted into the top of the frame, with a red arrow indicating the direction of movement.

Figure 3.6: Motion System CAD Assembly

3.6 Structural Frame Design

The structural frame provides mechanical support for the motion system, rotary platform, extrusion assembly, and control components.

- Support subsystem loads
- Minimize vibration
- Maintain alignment
- Provide adequate rigidity
- Facilitate assembly and maintenance

3.6.2 Material Selection

Several frame materials were considered including:

- Timber
- Aluminium extrusion
- Mild steel tubing

Mild steel square tubing was selected due to:

- Availability
- Ease of fabrication
- Good structural rigidity
- Welding compatibility
- Cost effectiveness

3.6.3 Structural Design

The frame geometry was developed to support the moving gantry system while providing sufficient space for the rotary platform and frosting operations.

The design incorporates:

- Welded base structure
- Gantry supports
- Equipment mounting interfaces
- Electrical enclosure mounting locations

Basic loading calculations were performed to verify that the frame could support the anticipated subsystem masses without excessive deflection.

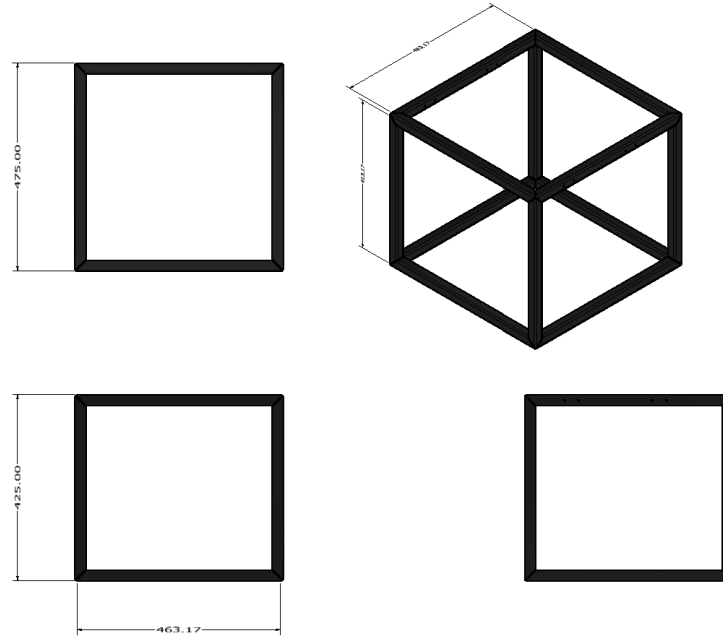


Figure 3.7: Structural Frame CAD Model

3.6.4 CAD Model

The complete frame was modelled prior to fabrication to verify dimensional compatibility with all machine subsystems.

3.7 Theta-Axis Rotary Platform Design

3.7.1 Design Requirements

The rotary platform was developed to support continuous rotation of the cake during frosting operations.

The platform was required to:

- Support expected cake loads
- Provide smooth rotation
- Allow speed control
- Integrate with machine control software
- Enable side-wall frosting operations

3.7.2 Rotary Platform Selection

Commercial rotary tables and industrial indexing systems were reviewed during the design process.

A custom rotary platform was selected because it:

- Reduced system cost
- Simplified integration
- Allowed customization of dimensions
- Supported project-specific requirements

3.7.3 Design Calculations

Rotary platform design calculations considered:

- Cake mass
- Platform loading
- Rotational speed requirements
- Motor torque requirements

The selected drive system was sized to provide stable rotational motion during frosting operations.

3.7.4 CAD Model

A complete CAD model was developed to evaluate:

- Platform dimensions
- Bearing arrangement
- Motor mounting
- Transmission layout

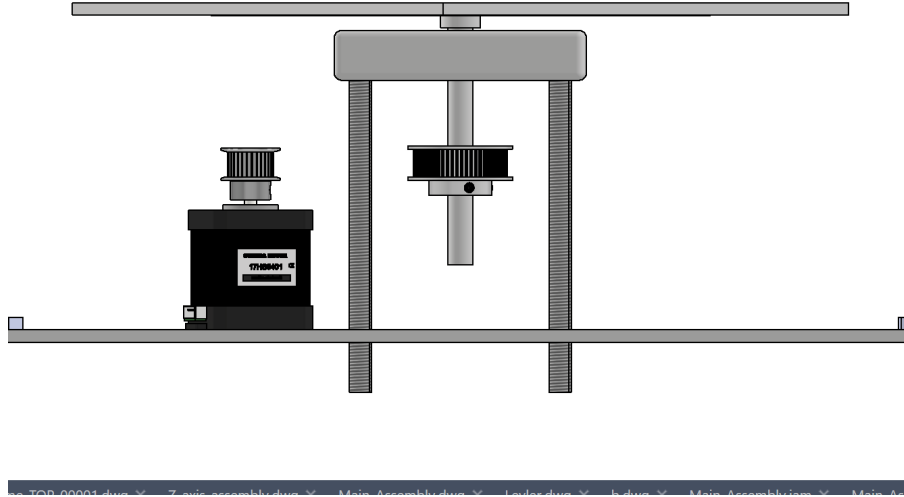


Figure 3.8: Rotary Platform CAD Model

3.8 Extrusion System Design

The extrusion system was developed to provide controlled deposition of buttercream frosting while overcoming the limitations identified in previous frosting machine designs. Particular emphasis was placed on generating sufficient extrusion force, maintaining consistent flow, and supporting synchronization with machine motion.

3.8.1 Previous System Limitations

The previous automated frosting machine developed within the department utilized:

- A hopper-based frosting reservoir,
- A rubber piston extrusion mechanism,
- Whipped cream as the frosting material.

Several limitations were observed during operation:

- Insufficient extrusion force,
- Nozzle clogging,

- Inconsistent frosting deposition,
- Poor synchronization with rotational motion,
- Inability to frost cake side walls effectively.

These limitations informed the redesign of the frosting delivery subsystem.

3.8.2 Selection of Syringe-Based Extrusion

Several extrusion approaches were considered:

A syringe-based extrusion system was selected due to its ability to generate controlled extrusion forces while maintaining consistent frosting delivery.

3.8.3 Reservoir Design

The frosting reservoir was designed using a cylindrical syringe arrangement capable of accommodating the required frosting volume for typical frosting operations.

Design considerations included:

- Frosting capacity,
- Refill accessibility,
- Cleaning requirements,
- Compatibility with the extrusion mechanism.

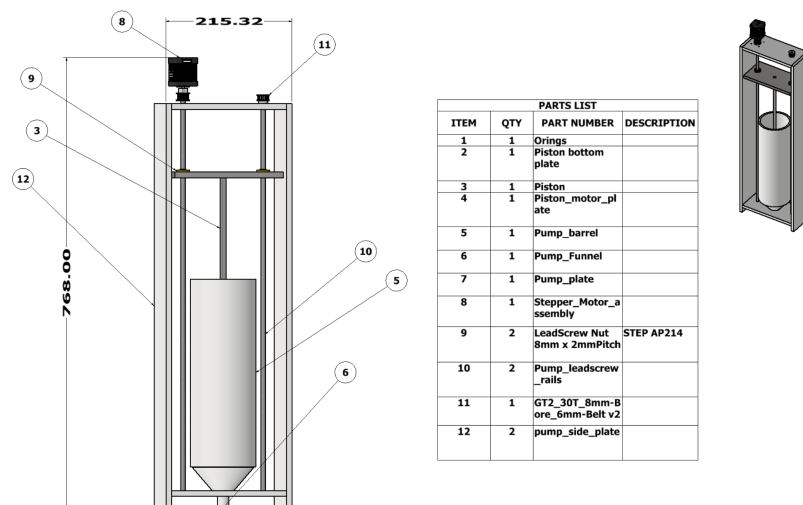


Figure 3.9: Syringe Reservoir Design

3.8.4 Lead Screw Drive Design

A lead screw mechanism was selected to convert motor rotation into linear plunger movement.

The lead screw system provides:

- high force generation,
- accurate linear positioning,
- controlled extrusion rate,
- compatibility with stepper motor control.

The extrusion rate can be regulated through the relationship between motor rotation and plunger displacement.

3.8.5 Extrusion Force Calculations

The extrusion system was sized to generate sufficient force for buttercream deposition through the selected nozzle geometry.

Calculations considered:

- frosting viscosity,
- nozzle diameter,
- plunger area,
- lead screw mechanical advantage,
- motor torque.

Detailed calculations are presented in Appendix X.

3.8.6 Plunger Design

A custom plunger assembly was designed to interface the lead screw mechanism with the frosting reservoir.

The design incorporates:

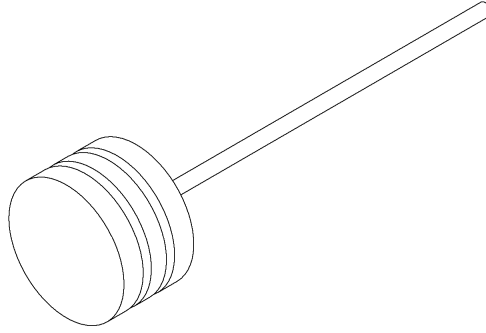


Figure 3.10: Custom Plunger Design

- lead screw attachment interface,
- guided linear motion,
- removable construction for cleaning,
- compatibility with additive manufacturing.

3.9 Frosting Nozzle Design

The nozzle forms the final interface between the extrusion system and the cake surface. Its geometry directly influences frosting flow behaviour, deposition consistency, and achievable frosting quality.

3.9.1 Design Requirements

The nozzle was required to:

- provide uniform frosting flow,
- minimize clogging,
- support buttercream deposition,
- enable repeatable frosting patterns,
- integrate with the syringe extrusion system.

3.9.2 Candidate Nozzle Designs

Two nozzle concepts were developed and evaluated during the design process.

The candidate designs differed in:

- outlet geometry,
- frosting distribution characteristics,
- manufacturing complexity,
- ease of cleaning.

Types of the proposed nozzle concepts are presented in Figure 3.11 and 3.12.



Figure 3.11: Round Nozzle

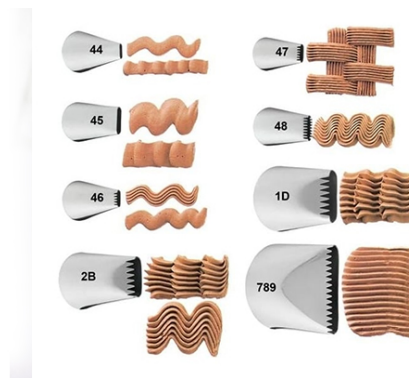


Figure 3.12: Basket_{Weave}Nozzle

Figure 3.14: Nozzle Designs

3.9.3 Selected Nozzle Design

The final nozzle design was selected based on:

- frosting flow consistency,
- manufacturability,
- integration requirements,
- observed extrusion performance.

The selected nozzle provided the most suitable balance between flow control and fabrication simplicity.

3.9.4 CAD Model and Fabricated Nozzle

The final nozzle geometry was modelled using CAD software and fabricated for system integration.

3.10 Control and Electrical System Design

The control and electrical system was developed to coordinate machine motion, frosting extrusion, operator interaction, and system monitoring.

3.10.1 Control Architecture

An ESP32-S3 microcontroller was selected as the primary controller.

The controller is responsible for:

- motion control,
- frosting control,
- rotary platform control,
- machine monitoring,
- web-based communication.

3.10.2 Motion Control Hardware

Motion control is achieved through stepper motor drivers connected to the ESP32 controller.

The motion subsystem consists of:

- X-axis drive,
- Y-axis drive,
- dual Z-axis drives,
- Theta-axis rotary drive.

Stepper motors were selected due to their positioning accuracy and compatibility with CNC-style control systems.

3.10.3 Extrusion Motor Control

Unlike the positioning axes, frosting extrusion is controlled independently.

The extrusion subsystem utilizes:

- DC motor,
- motor driver module,
- PWM speed control.

This arrangement enables real-time adjustment of frosting flow during automated frosting operations.

3.10.4 Power Distribution

The electrical system incorporates separate power domains for:

- motion control,
- logic circuitry,
- extrusion system,

- auxiliary devices.

This arrangement improves system reliability and reduces electrical interference between subsystems.

3.10.5 Control System PCB

A custom printed circuit board (PCB) was developed to simplify wiring and improve system integration.

The PCB provides:

- signal routing,
- power distribution,
- connector interfaces,
- subsystem integration.

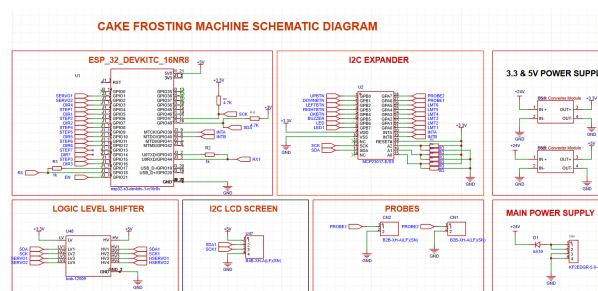


Figure 3.13: Controller Schematic Diagram

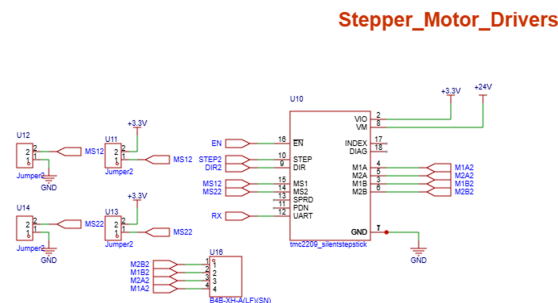


Figure 3.14: Motor_driverSchematicDiagram

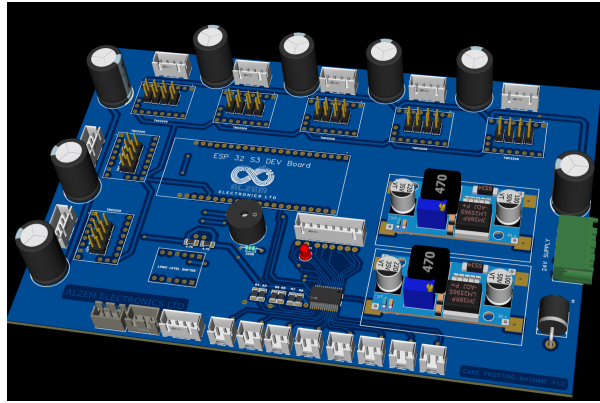


Figure 3.15: PCB layout

3.11 Custom Component Design

Several machine components were custom designed to satisfy dimensional, functional, and integration requirements that could not be achieved using commercially available components.

3.11.1 Shaft Couplers

Custom shaft couplers were developed to connect motors and transmission components possessing non-standard shaft dimensions.

Design objectives included:

- reliable torque transmission,
- ease of assembly,
- compatibility with available manufacturing methods.

3.11.2 Motion System Mounts

Custom mounting brackets were designed for:

- stepper motors,
- linear guides,
- bearing supports,

- gantry assemblies.

The designs were optimized to simplify assembly and maintain alignment.

3.11.3 Plunger Assembly

A custom plunger was developed to transfer force from the lead screw mechanism to the frosting reservoir.

The design incorporates:

- lead screw attachment features,
- guided motion,
- removable construction.

3.11.4 Nozzle Components

Custom nozzle interfaces and mounting components were designed to support integration with the extrusion assembly.

These components allow rapid installation and replacement of nozzle configurations during testing.

3.11.5 Control Enclosure and PCB Housing

Protective housings were designed to accommodate the control electronics, power distribution components, and custom PCB.

The enclosure provides:

- mechanical protection,
- cable management,
- maintenance accessibility,
- improved system organization.

The use of custom-designed components improved subsystem integration while reducing dependence on specialized commercially manufactured parts.

3.12 Fabrication Methodology

Following completion of the design phase, the machine components were fabricated using a combination of welding, machining, drilling, additive manufacturing, and printed circuit board fabrication techniques. The selected manufacturing methods were chosen based on component geometry, material properties, available facilities, and cost considerations.

3.12.1 Frame Fabrication

The structural frame was fabricated from mild steel square tubing. Individual frame members were cut to the required dimensions and joined using welding processes to produce a rigid support structure.

Following welding, the frame dimensions were verified to ensure proper alignment of the motion system and rotary platform mounting interfaces.



Figure 3.16: Welding



Figure 3.17: Grinding

3.12.2 Machining and Drilling Operations

Machining and drilling operations were performed to create mounting interfaces, fastening holes, shaft supports, and alignment features required for subsystem integration.

These operations were used for:

- motor mounting plates,
- bearing supports,
- rotary platform components,
- extrusion system interfaces,
- frame attachment points.



Figure 3.18: Turning

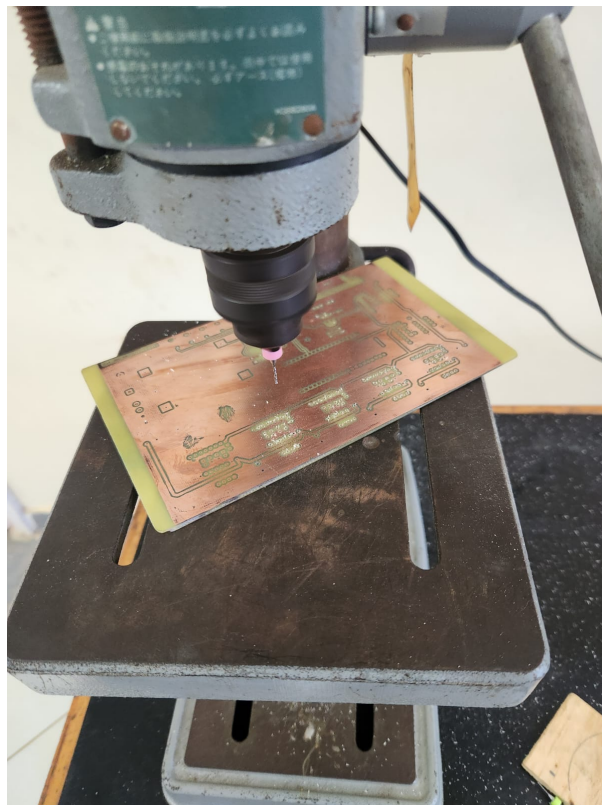


Figure 3.19: Drilling



Figure 3.20: Tapping

Figure 3.27: Machined and Drilled Components

3.12.3 Additive Manufacturing

Several custom components were produced using FDM 3D printing.

Components manufactured using additive manufacturing included:

- shaft couplers,
- motion system mounts,
- plunger components,
- nozzle prototypes,
- electronic enclosures.

The use of additive manufacturing enabled rapid prototyping and design iteration during development.

Figure 3.21: 3D Printed Components



Figure 3.21: 3D_{printing}

3.12.4 PCB Fabrication

A custom PCB was fabricated to simplify wiring and subsystem integration.

The fabrication process involved:

- schematic development,
- PCB layout generation,
- board manufacturing,
- component assembly,
- electrical verification.

The completed PCB was subsequently integrated into the control system.

3.13 Electrical Integration

Electrical integration involved the installation and interconnection of all control, power, sensing, and actuation components required for machine operation.

3.13.1 Wiring Architecture

A structured wiring approach was adopted to improve reliability, maintenance, and troubleshooting.

The wiring system consisted of:

- power distribution wiring,
- motion control wiring,
- communication wiring,
- peripheral device connections.

3.13.2 Motion System Integration

The X, Y, Z, and Theta-axis motors were connected to their respective drivers and integrated with the controller.

Electrical verification was performed to confirm:

- correct motor direction,
- driver operation,
- signal integrity,
- communication with the controller.

3.13.3 Extrusion System Integration

The extrusion motor and driver circuitry were installed and connected to the control system.

Integration included:

- PWM control connections,
- power connections,
- direction control signals,
- safety shutdown functionality.

3.13.4 Power Distribution Integration

Power distribution components were installed to supply electrical energy to:

- motion control hardware,
- logic circuitry,
- extrusion system,
- auxiliary components.

Voltage levels and current requirements were verified prior to commissioning.

3.13.5 Electrical Verification

Electrical testing was performed following integration to verify:

- continuity,
- polarity,
- power distribution,
- subsystem functionality.

The electrical system was verified before software deployment and system testing.

3.14 Embedded Software Development and Control Implementation

The automated frosting machine is controlled by embedded software running on an ESP32-S3 microcontroller. The software coordinates machine motion, frosting extrusion, rotary platform control, operator interaction, and automated frosting routines.

3.14.1 Software Architecture

The software was developed using a modular architecture consisting of:

- motion control functions,

- extrusion control functions,
- machine monitoring functions,
- web interface functions,
- automated frosting routines.

3.14.2 Cake Mapping Algorithms

Prior to frosting operations, the machine must determine the dimensions and location of the cake.

For circular cakes, a three-point mapping method was implemented in which three points on the cake perimeter are selected. The software then calculates the cake centre and radius.

For rectangular cakes, two diagonal corner points are selected to establish the frosting workspace.

The resulting dimensions are used during automated frosting operations.

3.14.3 Automated Frosting Algorithms

Dedicated algorithms were developed for automated frosting operations.

Implemented routines include:

- side-wall frosting,
- side-wall levelling,
- circular top frosting,
- rectangular top frosting.

The algorithms coordinate machine motion and frosting deposition to achieve consistent frosting coverage.

3.14.4 Dynamic Extrusion Control

To improve frosting consistency, dynamic extrusion control was implemented.

The extrusion system adjusts frosting delivery based on machine operating conditions to maintain a more uniform deposition rate.

Particular attention was given to frosting operations involving rotational motion, where frosting demand varies with changing deposition path lengths.

3.14.5 Dynamic Extrusion Control

To improve frosting consistency, dynamic extrusion control was implemented.

The extrusion system adjusts frosting delivery based on machine operating conditions to maintain a more uniform deposition rate.

Particular attention was given to frosting operations involving rotational motion, where frosting demand varies with changing deposition path lengths.

3.14.6 Text Rendering System

A vector-based text rendering system was developed to enable automated piping of text onto cake surfaces.

The system performs:

- text scaling,
- coordinate transformation,
- trajectory generation,
- extrusion control.

Retraction routines were incorporated to reduce frosting stringing between individual strokes.

3.14.7 Web Dashboard Development

A browser-based dashboard was developed to provide wireless machine control and monitoring.

The dashboard enables:

- manual jogging,

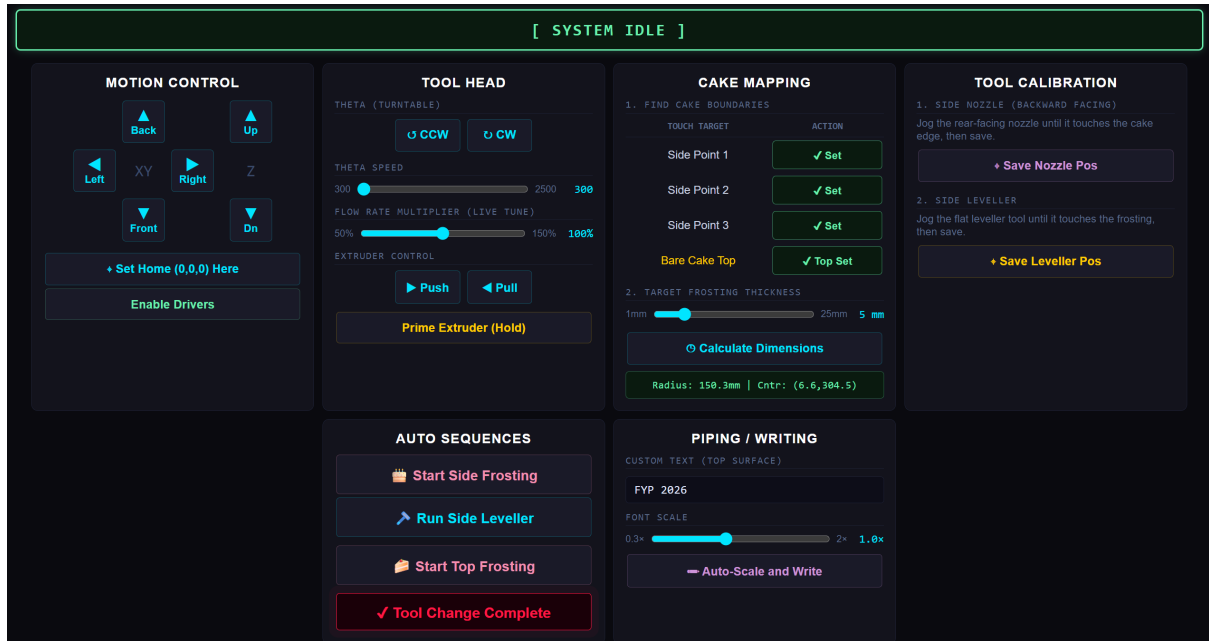


Figure 3.22: Web Dashboard

- machine setup,
- frosting routine selection,
- extrusion control,
- system monitoring.

The interface was implemented using:

- HTML,
- CSS,
- JavaScript.

Communication between the dashboard and controller is achieved through Wi-Fi networking.

3.15 System Integration and Testing

Following completion of mechanical, electrical, and software development activities, the machine was assembled and tested as an integrated system.

3.15.1 Mechanical Assembly

Mechanical assembly involved installation of:

- structural frame,
- motion system,
- rotary platform,
- extrusion subsystem,
- frosting nozzle assembly.

Subsystem alignment was verified prior to operation.

3.15.2 Functional Verification

Individual subsystems were tested to verify correct operation before integrated testing.

Subsystem verification included:

- motion testing,
- rotary platform testing,
- extrusion testing,
- communication testing,
- safety function testing.

3.15.3 Calibration Procedures

Calibration was performed to establish:

- axis travel accuracy,
- rotary platform operation,
- extrusion performance,
- frosting deposition consistency.

The calibration process ensured repeatable machine operation.

Chapter 4

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the results obtained from the fabrication, integration, and testing of the automated cake frosting machine. The completed machine was assembled from the mechanical, electrical, and software subsystems developed in Chapter 3 and subsequently evaluated through a series of subsystem and full-system tests. The chapter discusses the fabricated machine, subsystem performance, cake mapping verification, automated frosting operations, text piping capability, and overall system performance relative to the design requirements established during the design phase.

4.2 Fabricated Machine

The automated cake frosting machine was successfully fabricated and assembled. The completed system integrates a moving-gantry Cartesian motion system, a rotary cake platform, a syringe-based extrusion mechanism, an ESP32-based control system, and a web-enabled operator interface. The machine was designed to perform automated frosting deposition on both the top surface and side walls of cakes while supporting programmable frosting patterns and text piping operations.

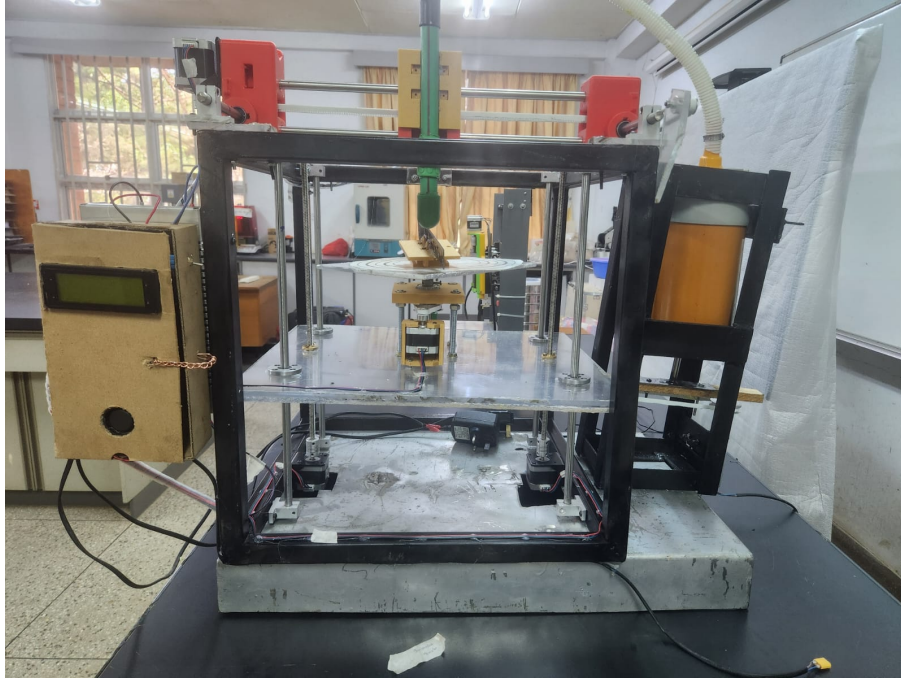


Figure 4.1: Completed Automated Cake Frosting Machine

4.2.1 Structural Frame

The structural frame was fabricated from mild steel square tubing and served as the primary support structure for all machine subsystems. The frame provided mounting interfaces for the gantry assembly, rotary platform, extrusion mechanism, and electrical components. Following fabrication and assembly, the frame exhibited adequate rigidity during machine operation with no significant structural deformation observed. The welded construction provided a stable platform for frosting operations and maintained alignment of the motion components.



Figure 4.2: Fabricated Structural Frame

4.2.2 Motion System

The motion system consisted of three Cartesian axes providing controlled movement of the frosting tools relative to the cake.

The X-axis and Y-axis provided horizontal positioning while the Z-axis controlled vertical movement of the gantry assembly. Stepper motors and lead screw mechanisms were used to achieve positional control. Following assembly, all axes completed their intended travel ranges and operated smoothly throughout the workspace. No significant binding or loss of motion was observed during testing.

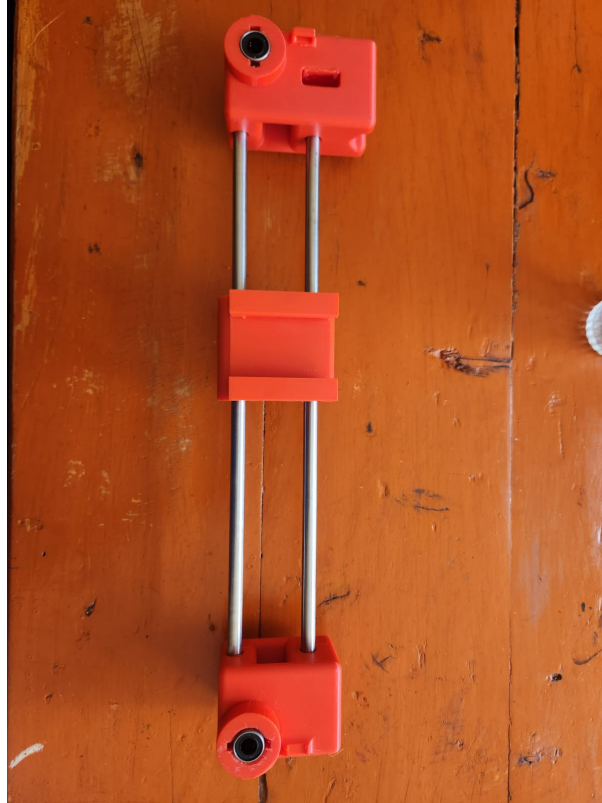


Figure 4.3: X-Axis Assembly

4.2.3 Rotary Platform

A custom rotary platform was developed to support automated side-wall frosting operations. The platform provided controlled rotational motion of the cake while maintaining stability throughout the frosting process.

The platform operated smoothly and enabled continuous rotation during frosting and levelling operations. Integration with the motion control system allowed synchronization between cake rotation and frosting deposition.

Figure 4.6: Rotary Platform Assembly

4.2.4 Extrusion System

The extrusion system consisted of a frosting reservoir, plunger mechanism, lead screw drive, and interchangeable frosting nozzles.

The fabricated system successfully generated sufficient force to extrude buttercream frosting through the selected nozzle configurations. The lead screw-driven plunger provided controlled frosting delivery and improved consistency compared to the hopper-based

system used in the previous project.



Figure 4.4: Extrusion System Assembly

4.2.5 Control System

The control system consisted of an ESP32-based controller, motor drivers, custom PCB, power distribution circuitry, and a wireless web dashboard.

The control hardware was integrated into the machine frame and provided coordinated control of motion, extrusion, and rotary platform operation. The web-based dashboard enabled machine monitoring, manual control, calibration, and execution of automated frosting routines.

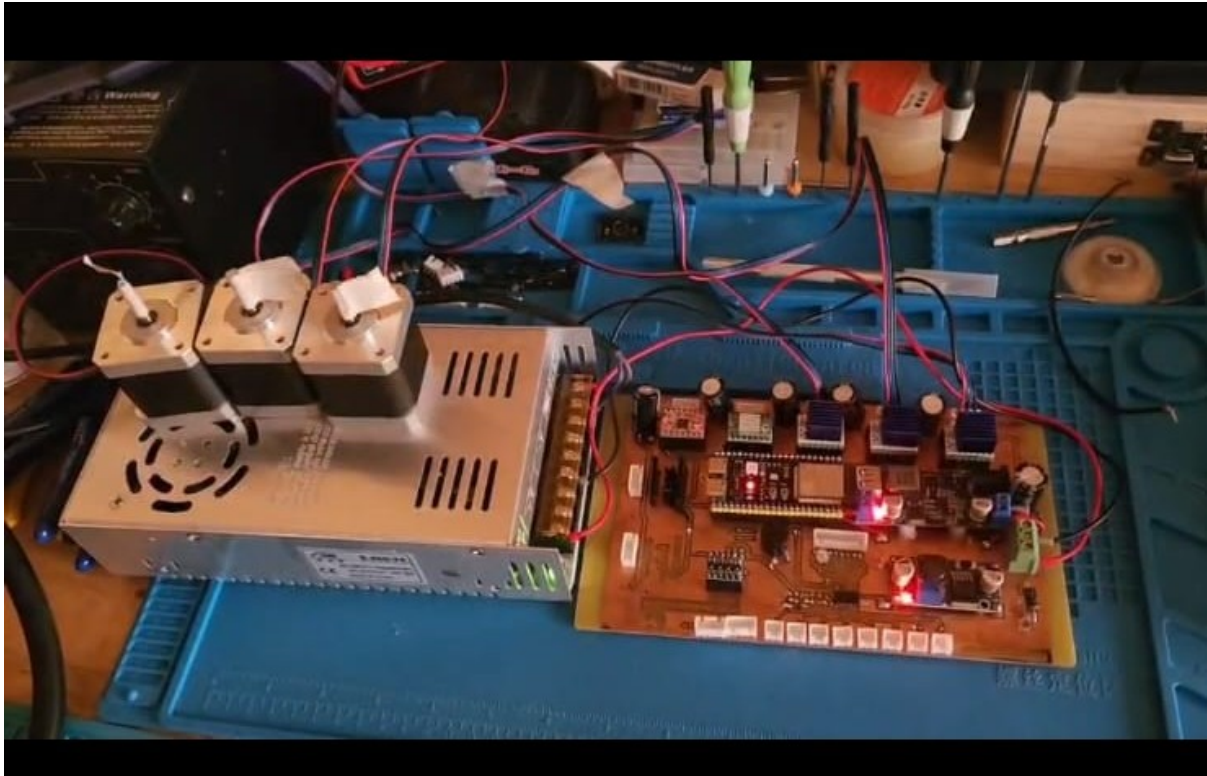


Figure 4.5: Control Electronics Assembly

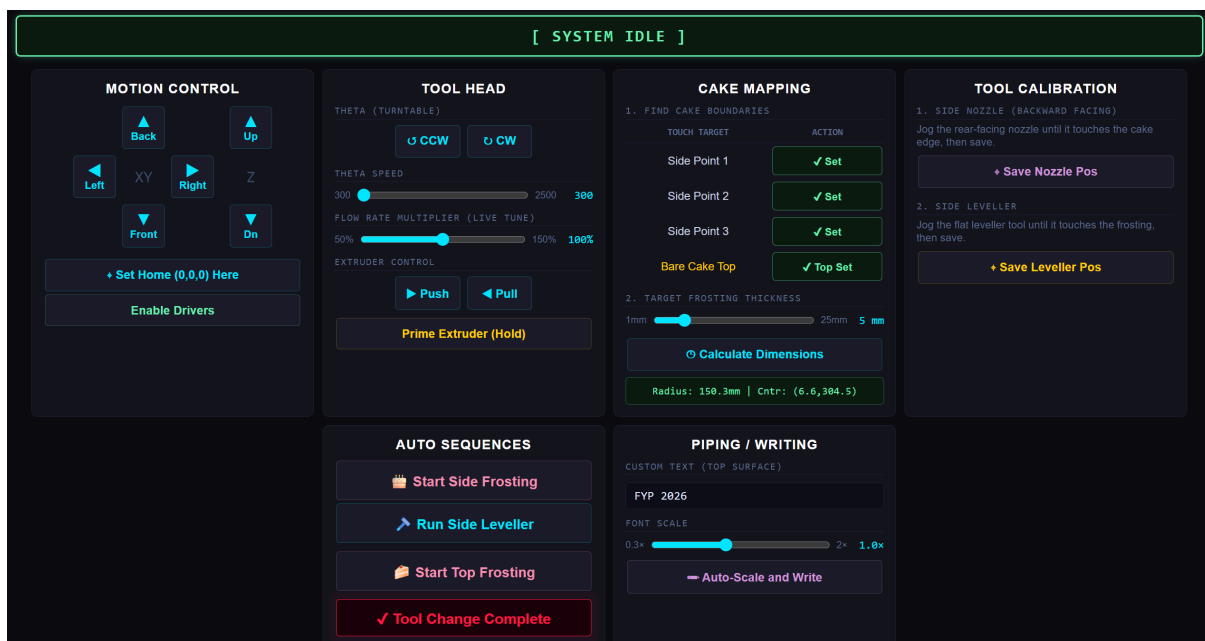


Figure 4.6: Web Dashboard Interface

4.3 Control System Verification

Following machine assembly, the control system was tested to verify communication between the operator interface and the machine hardware. Testing focused on dashboard accessibility, command execution, machine responsiveness, and subsystem actuation.

4.3.1 Controller Access and Dashboard Initialization

The ESP32 controller was configured as a wireless access point hosting the machine dashboard.

Testing involved:

- connecting to the controller network,
- accessing the web dashboard,
- verifying successful interface loading.

The dashboard loaded successfully and displayed the machine control interface, indicating successful communication between the operator device and the controller.

4.3.2 Dashboard Function Verification

Individual dashboard functions were tested to verify correct operation.

The following functions were evaluated:

- axis jogging,
- rotary platform control,
- extrusion control,
- emergency stop operation,
- parameter adjustment,
- machine status updates.

All functions responded correctly to operator commands.

4.3.3 Motion Command Verification

Motion commands were issued through the dashboard to verify control of the X, Y, and Z axes.

Each axis responded correctly to commanded movements and updated its position on the dashboard interface.

Testing confirmed:

- successful command transmission,
- correct axis response,
- real-time coordinate updates.

4.3.4 Rotary Platform Verification

The rotary platform was activated through the dashboard and tested at different operating speeds.

The platform responded correctly to control commands and maintained smooth rotational motion throughout testing.

4.3.5 Extrusion System Verification

The extrusion system was operated using the dashboard controls.

Forward extrusion, reverse extrusion, and stopping functions were tested to verify frosting delivery control.

The extrusion mechanism responded correctly to all commands and provided controllable frosting flow.

4.3.6 Integrated Command Execution

A complete operational sequence involving axis motion, rotary motion, and extrusion control was executed.

The machine successfully performed coordinated subsystem operation, demonstrating proper communication between the dashboard, controller, and machine hardware.

[Link to verification video](#)

4.4 Cake Mapping Verification

The automated frosting routines developed in this project require prior identification of the cake geometry. To achieve this, a cake mapping system was implemented to determine the dimensions and location of the cake within the machine workspace.

Separate mapping procedures were developed for circular and rectangular cakes. Verification testing was conducted to evaluate the ability of the system to correctly identify cake boundaries and generate the parameters required for automated frosting operations.

4.4.1 Circular Cake Mapping

Circular cake mapping was performed using the three-point circle method implemented within the machine software.

The operator manually positioned the frosting nozzle at three points along the perimeter of the cake using the dashboard controls. These coordinates were then processed by the controller to determine:

- cake centre coordinates,
- cake radius,
- frosting workspace.

The calculated centre and radius were displayed on the dashboard and subsequently used for automated side frosting, top frosting, and text placement operations.

Testing demonstrated that the mapping algorithm successfully identified the circular cake geometry and generated the required parameters for subsequent frosting operations.

4.4.2 Rectangular Cake Mapping

Rectangular cake mapping was performed by selecting two diagonally opposite corners of the cake.

Using the recorded coordinates, the controller generated a rectangular workspace defining the frosting area.

The calculated workspace provided:

- minimum X coordinate,

- maximum X coordinate,
- minimum Y coordinate,
- maximum Y coordinate.

These dimensions were subsequently used for raster-based frosting operations and automated text placement.

Testing confirmed successful identification of the rectangular cake boundaries and accurate generation of the frosting workspace.

4.4.3 Mapping System Evaluation

The implemented mapping system successfully generated the geometric parameters required for automated frosting operations without requiring physical templates or external measurement devices.

The calculated dimensions were used directly by the frosting algorithms, enabling automatic generation of frosting paths for both circular and rectangular cakes.

The results demonstrate that the mapping approach provides a practical and flexible method for adapting machine operation to cakes of different sizes and shapes.

4.5 Automated Frosting Tests

Following successful machine integration and cake mapping verification, automated frosting trials were conducted to evaluate the ability of the machine to perform frosting operations using the developed control algorithms.

Due to project time constraints, experimental validation was limited to circular top-surface frosting using buttercream frosting. The implemented frosting routine utilized the cake mapping system, synchronized rotary motion, and controlled extrusion to generate a continuous spiral frosting pattern.

The remaining automated routines, including side-wall frosting, side levelling, rectangular frosting, and automated text piping, were successfully implemented in software but were not experimentally evaluated within the project period.

4.5.1 Circular Top Frosting

Circular top frosting was selected as the primary proof-of-concept test because it required simultaneous operation of the major machine subsystems, including cake mapping, rotary positioning, motion control, and frosting extrusion.

The frosting routine was initiated after successful cake mapping. The nozzle was positioned at the outer cake radius while the rotary platform rotated continuously. Simultaneously, the nozzle moved progressively towards the centre of the cake, generating a spiral frosting pattern across the cake surface.

link to top frosting video

The frosting operation was completed successfully, demonstrating coordinated control of the motion system, rotary platform, and extrusion mechanism.

Observations recorded during testing included:

- successful execution of the generated frosting path,
- continuous frosting deposition,
- effective synchronization between cake rotation and nozzle movement,
- adequate frosting coverage of the cake surface,
- stable machine operation throughout the frosting process.

The results confirmed the feasibility of the proposed machine architecture for automated cake frosting applications.

4.5.2 Future Frosting Operations

Additional frosting routines were developed as part of the machine software architecture but were not experimentally validated during the project period.

These include:

- side-wall frosting,
- side levelling,
- rectangular top frosting,

- automated text piping.

The implemented algorithms remain available for future testing and performance evaluation. Their successful implementation within the control software provides a foundation for further development and expanded machine capability.

4.6 Text Piping Capability

A text piping system was developed to enable the automated deposition of letters and simple decorative text onto cake surfaces.

The implemented system converts vector stroke definitions into machine motion commands, allowing the frosting nozzle to trace characters using coordinated X-Y motion and controlled frosting extrusion.

Additional software features were incorporated including:

- automatic text scaling,
- coordinate transformation,
- frosting retraction,
- Z-axis lift movements between character segments.

The text piping functionality was successfully implemented within the machine software and integrated into the web dashboard interface.

However, due to project time constraints, experimental frosting trials involving text deposition were not conducted during the project period.

Future work will focus on evaluating:

- text readability,
- dimensional accuracy,
- frosting stringing behaviour,
- scaling performance on different cake sizes.

4.7 Performance Evaluation Against Design Requirements

The completed machine was evaluated against the engineering requirements established in Section 3.3.4.

Table 4.5: Evaluation Against Design Requirements

The evaluation indicates that the primary project objectives were achieved. The completed system demonstrated successful integration of the mechanical, electrical, and software subsystems required for automated cake frosting.

4.8 Comparison with Previous Project

A comparison was conducted between the previously developed cake frosting machine and the machine developed in the present project.

The previous system utilized an X-Z-Rotary architecture together with a hopper-based extrusion mechanism operating with whipped cream. Several operational challenges were reported, including insufficient extrusion force, clogging, frosting inconsistency, and difficulties achieving synchronized motion.

The proposed machine was developed specifically to address these limitations.

Table 4.6: Comparison with Previous Project

The redesign introduced several improvements, particularly in extrusion control, machine flexibility, software capability, and overall system integration.

The successful frosting trials demonstrated that the proposed architecture addressed many of the limitations observed in the earlier system while providing a platform for future expansion and refinement.

4.9 Challenges Encountered During Development

Several challenges were encountered during the design, fabrication, integration, and testing of the automated cake frosting machine. These challenges influenced subsystem development and resulted in iterative modifications throughout the project.

4.9.1 Frosting Behaviour and Consistency

Buttercream frosting exhibited variations in consistency depending on preparation conditions and ambient temperature. Changes in frosting consistency affected extrusion performance and occasionally required adjustment of extrusion parameters to maintain uniform flow.

4.9.2 Extrusion System Tuning

Achieving consistent frosting delivery required repeated adjustment of nozzle dimensions, plunger movement rates, and extrusion motor settings. Excessive extrusion resulted in frosting accumulation, while insufficient extrusion reduced frosting coverage.

4.9.3 Motion Synchronization

The frosting process required coordination between gantry motion, rotary platform movement, and extrusion control. Developing synchronized operation between these subsystems required extensive software testing and parameter tuning.

4.9.4 Mechanical Alignment and Fabrication Tolerances

Minor alignment errors were encountered during assembly of the motion system. Fabrication tolerances introduced small positioning deviations that required mechanical adjustment and calibration before satisfactory operation was achieved.

4.9.5 Time Constraints

Although the machine was successfully fabricated and demonstrated automated circular frosting, limited project time restricted the extent of experimental testing. Consequently, side-wall frosting, levelling operations, and text piping functions were implemented but not fully validated through practical frosting trials.

Despite these challenges, the machine was successfully integrated and demonstrated the core functionality required for automated cake frosting.

4.10 Chapter Summary

This chapter presented the fabrication results, subsystem verification, cake mapping validation, automated frosting tests, and overall performance evaluation of the automated cake frosting machine.

The completed machine successfully integrated a Cartesian gantry motion system, rotary cake platform, syringe-based extrusion mechanism, ESP32-based control system, and web-enabled operator interface. Control system verification confirmed successful communication between the dashboard and machine hardware, while cake mapping tests demonstrated the ability to automatically determine cake geometry for frosting operations.

Automated frosting trials successfully demonstrated circular top-surface frosting using buttercream frosting. The results confirmed effective synchronization between motion control, cake rotation, and frosting extrusion. Evaluation against the design requirements indicated that the major project objectives were achieved.

Comparison with the previous project showed improvements in extrusion capability, control flexibility, software functionality, and overall machine integration. Although some advanced functions were not experimentally validated within the project period, the implemented hardware and software architecture provides a foundation for future development and testing.

The next chapter presents the conclusions drawn from the project and provides recommendations for future work.

Chapter 5

CHAPTER 5

CONCLUSIONS AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the conclusions drawn from the design, fabrication, and testing of the automated cake frosting machine. The chapter also evaluates the achievement of the project objectives, highlights the major contributions of the work, and provides recommendations for future development and improvement of the system.

5.2 Conclusion

The aim of this project was to design and fabricate an automated cake frosting machine capable of applying frosting with minimal operator intervention while improving consistency, repeatability, and process automation.

A requirement-driven design methodology was adopted, beginning with the selection and analysis of buttercream frosting as the target material. The engineering requirements derived from the frosting characteristics guided the development of the machine architecture, extrusion system, motion system, control system, and software algorithms.

The completed machine successfully integrated a moving-gantry Cartesian motion system, a rotary cake platform, a syringe-based extrusion mechanism, an ESP32-based controller, and a wireless web-based operator interface. The developed control software incorporated cake mapping algorithms, automated frosting routines, and programmable motion control capabilities.

Testing demonstrated successful communication between the operator interface and machine hardware, reliable motion control, effective cake mapping, and automated circular top-surface frosting using buttercream frosting. The results confirmed that the proposed machine architecture is capable of performing automated frosting operations

while maintaining coordinated motion and controlled material deposition.

The project further demonstrated that a syringe-based extrusion mechanism provides improved control when handling higher-viscosity frosting materials compared to the hopper-based approach used in previous work. The integration of rotary motion and programmable control also established a foundation for future side-wall frosting, levelling, and decorative piping operations.

Overall, the project achieved its primary objective of developing a functional automated cake frosting machine and demonstrated the feasibility of applying CNC-based automation techniques to cake decoration processes.

5.3 Achievement of Objectives

The objectives established at the beginning of the project were evaluated based on the final machine performance.

Objective 1: To design an automated cake frosting machine

This objective was achieved through the development of a complete machine architecture consisting of mechanical, electrical, and software subsystems. CAD models were developed and used throughout the design process.

Objective 2: To fabricate and integrate the machine components

The machine was successfully fabricated and assembled. The motion system, rotary platform, extrusion mechanism, control electronics, and supporting structure were integrated into a functional system.

Objective 3: To develop a control system for automated frosting operations

An ESP32-based control system and web-enabled dashboard were successfully developed. The system provided machine control, cake mapping functionality, and execution of automated frosting routines.

Objective 4: To evaluate the performance of the machine

Subsystem testing, cake mapping verification, and automated frosting trials were conducted. The machine successfully demonstrated automated circular top frosting using buttercream frosting.

The project objectives were therefore successfully achieved.

5.4 Contributions of the Project

The project made several contributions to the development of automated cake decoration systems.

The major contributions include:

- Development of an automated cake frosting machine based on a moving-gantry Cartesian architecture.
- Implementation of a syringe-based extrusion system capable of handling buttercream frosting.
- Development of a rotary cake platform for synchronized frosting operations.
- Implementation of circular and rectangular cake mapping algorithms.
- Development of automated frosting control algorithms.
- Development of a web-enabled wireless operator interface.
- Implementation of a vector-based text piping framework for future decorative applications.

The project also addressed several limitations identified in previous work, particularly those related to extrusion force generation, frosting consistency, and machine flexibility.

5.5 Recommendations for Future Work

Although the developed machine successfully demonstrated automated frosting functionality, several opportunities exist for further improvement.

Side-Wall Frosting Validation

The side-wall frosting and levelling routines should be experimentally evaluated under different frosting conditions and cake sizes to quantify performance and frosting quality.

Text Piping Evaluation

The implemented text piping functionality should be tested using frosting materials to evaluate readability, dimensional accuracy, and decorative quality.

Closed-Loop Control

Future versions of the machine may incorporate position feedback devices and sensors to improve motion accuracy and enable closed-loop control.

Advanced Extrusion Control

Additional research may be conducted on adaptive extrusion control techniques that automatically compensate for variations in frosting consistency and environmental conditions.

Food-Grade Construction

Future designs should incorporate food-grade materials and removable components to improve cleaning, hygiene, and suitability for commercial food production environments.

Machine Vision Integration

Camera-based vision systems may be incorporated to automate cake detection, alignment, and quality inspection processes.

Decorative Pattern Generation

Future software development could enable the generation of more complex decorative patterns, graphics, and custom designs directly from user inputs.

5.6 Final Remarks

Automation continues to transform manufacturing and food processing industries through improved productivity, consistency, and process control. This project demonstrates that similar principles can be successfully applied to cake decoration through the integration of mechanical systems, embedded control, and software automation.

The developed automated cake frosting machine provides a foundation for future research and development in automated cake decoration systems. While further refinement and testing remain necessary, the successful design, fabrication, and demonstration of the machine confirm the feasibility of automated frosting operations using buttercream frosting and programmable CNC-based control techniques.

References

- [1] Fernandes, R. R., Calver, K. E., Hayes, D., Zorrilla, S. E., & Wilson, D. I. (2024). Aeration and rheology of buttercream icings. *Journal of Food Engineering*, 364, 111775.
- [2] Meza, B. E., Peralta, J. M., & Zorrilla, S. E. (2021). Rheological characterization of buttercream icing. *International Journal of Food Properties*, 24(1), 123-135.
- [3] Kumbár, V., Nedomová, Š., & Buchar, J. (2022). Use of rheological plastic models to describe the flow behaviour of unconventional chocolate masses. *Journal of Food Science and Technology*, 59(4), 1450-1460.
- [4] FluidNC Team. (2026). *FluidNC Firmware Documentation: High-Performance Grbl-Compatible Motion Control for ESP32*. Available at: <https://github.com/bdring/FluidNC>
- [5] Herschel, W. H., & Bulkley, R. (1926). Konsistenzmessungen von Gummi-Benzollösungen. *Kolloid-Zeitschrift*, 39(4), 291-300.
- [6] Zhang, M., & Bhandari, B. (2022). *Fundamentals of 3D Food Printing and Applications*. Academic Press.
- [7] J. Sun, Z. Peng, W. Zhou, J. Y. Fuh, G. S. Hong, and A. Chiu, “A review on 3D printing for customized food fabrication,” *Procedia Manufacturing*, vol. 1, pp. 308–319, 2015.
- [8] J. I. Lipton, M. Cutler, F. Nigl, D. Cohen, and H. Lipson, “Additive manufacturing for the food industry,” *Trends in Food Science & Technology*, vol. 43, no. 1, pp. 114–123, 2015.
- [9] L. Hao, S. Mellor, O. Seaman, J. Henderson, N. Sewell, and M. Sloan, “Material characterisation and process development for chocolate additive layer manufacturing,” *Virtual and Physical Prototyping*, vol. 5, no. 2, pp. 57–64, 2010.

- [10] D. Periard, N. Schaal, M. Schaal, E. Malone, and H. Lipson, "Printing food: the 3D extrusion of culinary ingredients," *3D Printing and Additive Manufacturing*, vol. 2, no. 1, pp. 1–9, 2015.
- [11] C. Severini and D. Derossi, "Could the 3D printing technology be a useful strategy to obtain customized nutrition?" *Journal of Clinical Gastroenterology*, vol. 50, pp. S175–S178, 2016.
- [12] S. Mantihal, C. Prakash, and S. Bhandari, "Textural modification of 3D printed dark chocolate by varying internal infill structure," *Food Research International*, vol. 104, pp. 104–112, 2017.
- [13] H. W. Kim, I. J. Bae, and S. Park, "Classification of the rheological properties of food materials for 3D printing: a review," *Journal of Food Engineering*, vol. 215, pp. 1–11, 2018.
- [14] F. Yang, M. Zhang, B. Bhandari, and Y. Liu, "Investigation on lemon juice gel as 3D printing ink for food manufacturing," *LWT - Food Science and Technology*, vol. 87, pp. 67–76, 2018.
- [15] D. L. Cohen, J. I. Lipton, M. Cutler, D. Coulter, A. Vesco, and H. Lipson, "Hydrocolloid printing: A novel platform for customized food production," in *Solid Freeform Fabrication Symposium*, Austin, TX, 2009.
- [16] Z. Liu, M. Zhang, B. Bhandari, and Y. Wang, "3D printing: Printing precision and application in food sector," *Trends in Food Science & Technology*, vol. 69, pp. 83–94, 2017.
- [17] V. Vancauwenberghe, E. Katalagarianakis, N. Baelmans, et al., "3D printing of plant tissue for personalized food," *Journal of Food Engineering*, vol. 200, pp. 19–28, 2017.
- [18] J. F. Steffe, *Rheological Methods in Food Process Engineering*. 2nd ed., East Lansing, MI: Freeman Press, 1996.
- [19] S. D. Holdsworth, "Rheological models used for the prediction of the flow properties of food products: a literature review," *Transactions of the Institution of Chemical Engineers*, vol. 71, no. C3, pp. 139–179, 1993.

- [20] M. P. Groover, *Automation, Production Systems, and Computer-Integrated Manufacturing*. 2nd ed., Prentice Hall, 2001.
- [21] Y. Koren, *The Global Manufacturing Revolution: Product-Process-Business Integration and Reconfigurable Systems*. Wiley, 2010.

Appendices

Appendix A: Detailed Mechanical Drawings

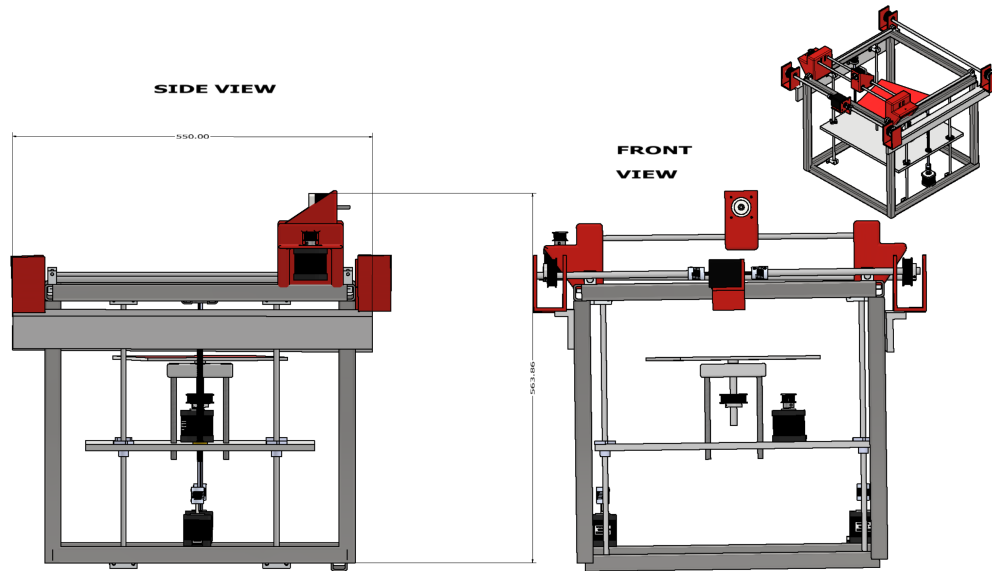


Figure 5.1: Engineering Orthographic Views: Overall Chassis Dimensions (470mm $\times$ 475mm $\times$ 580mm).

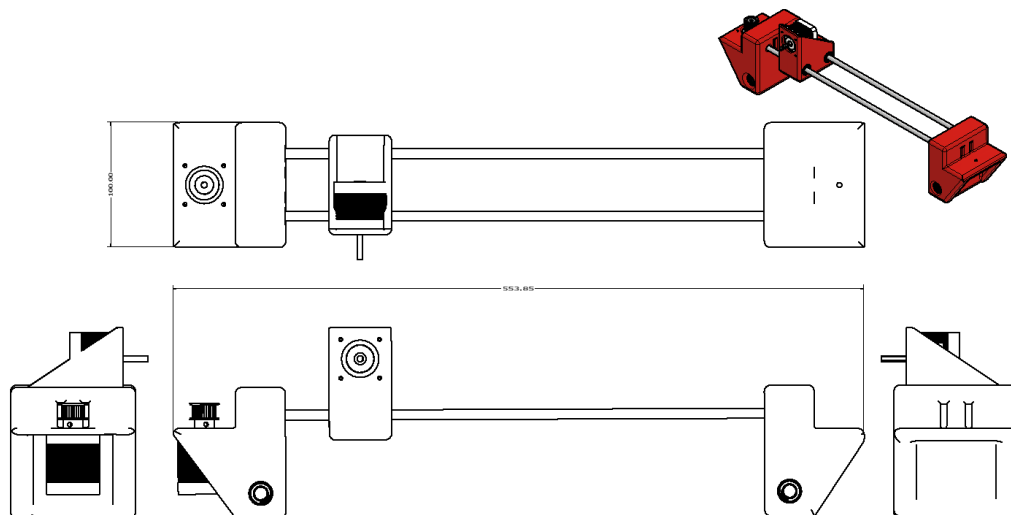


Figure 5.2: X-Axis Carriage Assembly: Details of the Vertically Stacked 8mm Rod Configuration.

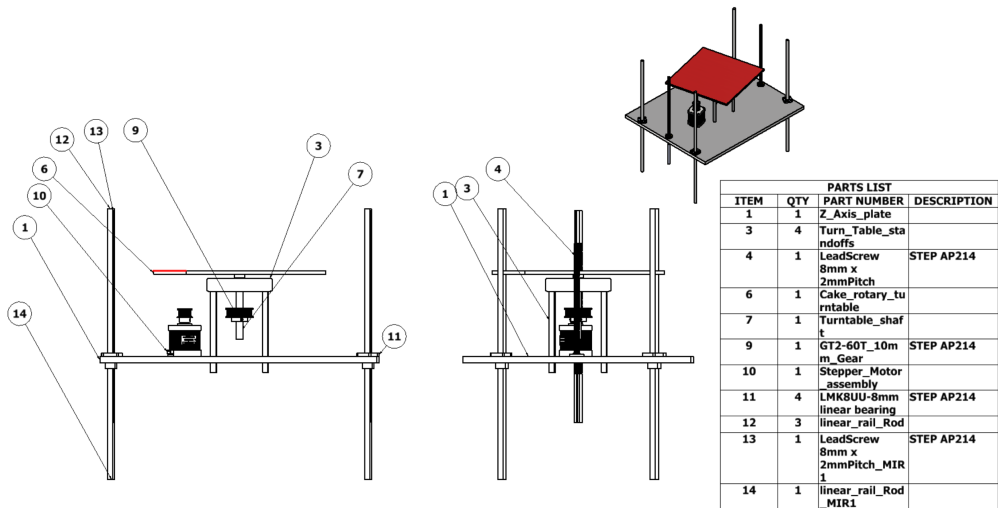


Figure 5.3: Z-Axis Lifting Bed: Independent Lead Screw Synchronization and 4-Corner Constraint System.

Appendix B: Finalized Project Budget and Bill of Materials

Table 5.1: Consolidated Bill of Materials for the 5-Axis CNC Platform

Item	Description	Qty	Unit Price	Total (Kes)
NEMA 17 Motors	1.8° High Torque (X, Y, Z, Theta)	5	1,800	9,000
NEMA 23 Motor	2.2 Nm Holding Torque (Extruder)	1	4,500	4,500
ESP32-S3 Kit	DevKitC N16R8 (240 MHz)	1	2,500	2,500
T8 Lead Screws	8mm Lead, 500mm Length	2	1,500	3,000
Chrome Rods	8mm/10mm Diameter, Case-Hardened	6	1,000	6,000
Mild Steel Tube	25mm x 25mm Square Section	1	2,500	2,500
Power Supply	24V 15A SMPS (360W)	1	3,500	3,500
Piston Cylinder	1L Positive Displacement Assembly	1	3,500	3,500
TOTAL				34,500

Appendix C: System Electrical Schematic

The following schematic details the signal flow and power domain separation implemented to ensure 32-bit controller stability during high-torque actuation.

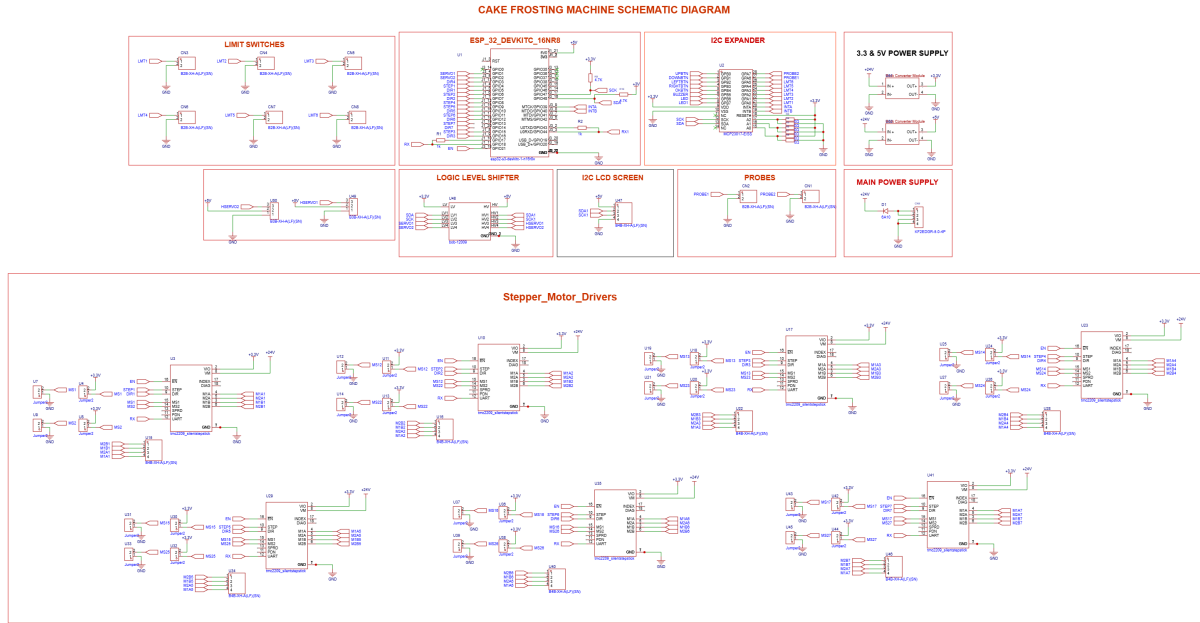


Figure 5.4: Complete Electrical Interconnect Diagram: ESP32-S3, TMC2209 Drivers, and Buck Converters.

Appendix D: Production and Assembly Workflow

The fabrication and validation of the system followed a structured 10-week technical workflow:

1. **Structural Fabrication (Weeks 1-2):** Cutting and welding of the mild steel chassis; alignment of the Y-axis support rails.
2. **Axis Integration (Weeks 3-4):** Installation of the overhead Flying Gantry (X-Y) and the dual Z-axis lifting bed.
3. **Extrusion Subsystem (Weeks 5-6):** Assembly of the 1L piston pump and the remote NEMA 23 drive interface.
4. **Electrical Population (Weeks 7-8):** Population of the control board, wiring of the 24V power distribution, and verification of the buck converter outputs.
5. **Firmware and Calibration (Weeks 9-10):** Configuration of the FluidNC ‘pins’ and ‘axes’ parameters; implementation of the Herschel-Bulkley retraction macros and batch-specific flow multiplier (Φ) calibration.